

Euclid's Elements

Modernized Unicode Edition

Compiled from Fitzpatrick Edition

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Contents

ELEMENTS BOOK 1

Fundamentals of Plane Geometry Involving Straight-Lines

κ/εντρῷ μτ=ω α διαστ/ηματι δτ=ω αβ κ/υκλοξ γεγρ/αφθω (ο βγδ, καὶ π/αλιν κ/εντρῷ μτ=ω β διαστ/ηματι δτ=ω βα κ/υκλοξ γεγρ/αφθω (ο αγε, καὶ)απτο=υ γ σημείου, καθ) (τ/εμνουσιν)αλλ/ηλουξ ὅι κ/υκλοι,)επ/ι τα, β σημει=ια)επεξέυθθωσαν έυθε=ιαι αὶ γα, γβ.

καὶ)επεί τα σημει=ιον κ/εντρον)εσtto=υ γδβ κ/υκλου,)/ιση)εστ(η αγ τ=η αβ; π/αλιν,)επεί τβ σημει=ιον κ/εντρον)εσtto=υ γαε κ/υκλου,)/ιση)εστ(η βγ τ=η βα.)εδείθθη δκαὶ (η γα τ=η αβ)/ιση; (εκατ/ερα)/αρα τ=ων γα, γβ τ=η αβ)εστιν)/ιση. τδτ=ω α)υτ=ω)/ισα καὶ)αλλ/ηλοιξ)εστ)/ισα; καὶ (η γα)/αρα τ=η γβ)εστιν)/ιση; αὶ τρε=ιξ)/αρα αὶ γα, αβ, γβ)/ισαι)αλλ/ηλαιξ έις/ιν.

)ις/οπλευρον)/αρα)εστταβγ τρ/ιγωνον. καὶ συν/εσταται)επτ=ηξ δοθείσηξ έυθείαξ πεπερασμ/ενηξ τ=ηξ αβ. (/οπερ)/εδει ποιησαι.

Proposition 1

To construct an equilateral triangle on a given finite straight-line.

Let AB be the given finite straight-line.

So it is required to construct an equilateral triangle on the straight-line AB .

Let the circle BCD with center A and radius AB be drawn[†] [Post. 3], and again let the circle ACE with center B and radius BA be drawn [Post. 3]. And let the straight-lines CA and CB be joined from the point C , where the circles cut one another,[‡] to the points A and B (respectively) [Post. 1].

And since the point A is the center of the circle CDB , AC is equal to AB [Def. 1.15]. Again, since the point B is the center of the circle CAE , BC is equal to BA [Def. 1.15]. But CA was also shown (to be) equal to AB . Thus, CA and CB are each equal to AB . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, CA is also equal to CB . Thus, the three (straight-lines) CA , AB , and BC are equal to one another.

Thus, the triangle ABC is equilateral, and has been constructed on the given finite straight-line AB . (Which is) the very thing it was required to do. [†] Literally, "have been drawn". The use of this peculiar Greek idiom will pass without further comment.

[‡] The assumption that the circles do indeed cut one another should be counted as an additional postulate. There is also an implicit assumption that two straight-lines cannot share a common segment.

2.

πρτ=ω δοθ/εντι σημείω τ=η δοθείση έυθεία)/ισην έυθε=ιαν θ/εσθαι.

)/εστω τμδοθσημε=ιον τα, (η δδοθε=ισα έυθε=ια (η βγ; δε=ι δπρτ=ω α σημείω τ=η δοθείση έυθεία τ=η βγ)/ισην έυθε=ιαν θ/εσθαι.

)επεξέυθθω γ)απτο=υ α σημείου)επ/ι τβ σημει=ιον έυθε=ια (η αβ, καὶ συνεστ/ατω)επ) α)υτ=ηξ τρ/ιγωνον)ις/οπλευρον τδαβ, καὶ)εκβεβλ/ησθωσαν)επ) έυθείαξ τ̂αιξ δα, δβ έυθε=ιαι αὶ αε, βζ, καὶ κ/εντρῷ μτ=ω β διαστ/ηματι δτ=ω βγ κ/υκλοξ γεγρ/αφθω (ο γηθ, καὶ π/αλιν κ/εντρῷ τ=ω δ καὶ διαστ/ηματι τ=ω δη κ/υκλοξ γεγρ/αφθω (ο ηκλ.

)επεί οὖν τῷ σημείῳ κέντρον ἔστω γηθ,)/ιση)εστ(η βγ τ=η βη. π/άλιν,)επεί τῷ σημείῳ κέντρον ἔστω ηκλ κ/υκλου,)/ιση)εστ(η δλ τ=η δη, (=ων (η δα τ=η δβ)/ιση)εστ/ιν. λοιπ)/αρά (η αλ λοιπ=η τ=η βη)εστιν)/ιση.)εδείθθη δὲ καὶ (η βγ τ=η βη)/ιση; (εκατ/ερα)/αρά τ=ων αλ, βγ τ=η βη)εστιν)/ιση. τδτ=ω α)υτ=ω)/ισα καὶ)αλλ/ηλοιξ)εστ)/ισα; καὶ (η αλ)/αρά τ=η βγ)εστιν)/ιση.

πρ)/αρά τ=ω δοθ/εντι σημείῳ τ=ω α τ=η δοθείσῃ ἐυθείᾳ τ=η βγ)/ιση ἐυθεῖα κεῖται (η αλ; (/οπερ)/εδει ποῖησαι.

Proposition 2[†]

To place a straight-line equal to a given straight-line at a given point (as an extremity).

Let A be the given point, and BC the given straight-line. So it is required to place a straight-line at point A equal to the given straight-line BC .

For let the straight-line AB be joined from point A to point B [Post. 1], and let the equilateral triangle DAB be constructed upon it [Prop. 1.1]. And let the straight-lines AE and BF be produced in a straight-line with DA and DB (respectively) [Post. 2]. And let the circle CGH with center B and radius BC be drawn [Post. 3], and again let the circle GKL with center D and radius DG be drawn [Post. 3].

Therefore, since the point B is the center of (the circle) CGH , BC is equal to BG [Def. 1.15]. Again, since the point D is the center of the circle GKL , DL is equal to DG [Def. 1.15]. And within these, DA is equal to DB . Thus, the remainder AL is equal to the remainder BG [C.N. 3]. But BC was also shown (to be) equal to BG . Thus, AL and BC are each equal to BG . But things equal to the same thing are also equal to one another [C.N. 1]. Thus, AL is also equal to BC .

Thus, the straight-line AL , equal to the given straight-line BC , has been placed at the given point A . (Which is) the very thing it was required to do. [†] This proposition admits of a number of different cases, depending on the relative positions of the point A and the line BC . In such situations, Euclid invariably only considers one particular case—usually, the most difficult—and leaves the remaining cases as exercises for the reader.

3.

δ/υο δοθείς=ων ἐυθεΐων)αν/ισων)απτ=ηξ μείζονοξ τ=η)ελ/ασσονι)/ισην ἐυθεῖαν)αφελε=ιν.

)εστωσαν αἱ δοθεῖσαι δ/υο ἐυθεῖαι)/ανισοι αἱ αβ, γ, (=ων μείζων)/εστω (η αβ; δε=ι δ)απτ=ηξ μείζονοξ τ=ηξ αβ τ=η)ελ/ασσονι τ=η γ)/ισην ἐυθεῖαν)αφελε=ιν.

κέισθω πρτ=ω α σημείῳ τ=η γ ἐυθείᾳ)/ιση (η αδ; καὶ κέντρῳ μτ=ω α διαστ/ηματι δτ=ω αδ κ/υκλοξ γεγρ/αφθω (ο δεξ.

καὶ)επεί τα σημείον κέντρον ἔστω δεξ κ/υκλου,)/ιση)εστ(η αε τ=η αδ;)αλλκαὶ (η γ τ=η αδ)εστιν)/ιση. (εκατ/ερα)/αρά τ=ων αε, γ τ=η αδ)εστιν)/ιση; (/ωστε καὶ (η αε τ=η γ)εστιν)/ιση.

δ/υο)/αρά δοθείς=ων ἐυθεΐων)αν/ισων τ=ων αβ, γ)απτ=ηξ μείζονοξ τ=ηξ αβ τ=η)ελ/ασσονι τ=η γ)/ιση)αφ/ηρηται (η αε; (/οπερ)/εδει ποῖησαι.

Proposition 3

For two given unequal straight-lines, to cut off from the greater a straight-line equal to the lesser.

Let AB and C be the two given unequal straight-lines, of which let the greater be AB . So it is required to cut off a straight-line equal to the lesser C from the greater AB .

Let the line AD , equal to the straight-line C , be placed at point A [Prop. 1.2]. And let the circle DEF be drawn with center A and radius AD [Post. 3].

And since point A is the center of circle DEF , AE is equal to AD [Def. 1.15]. But, C is also equal to AD . Thus, AE and C are each equal to AD . So AE is also equal to C [C.N. 1].

Thus, for two given unequal straight-lines, AB and C , the (straight-line) AE , equal to the lesser C , has been cut off from the greater AB . (Which is) the very thing it was required to do.

4.

ἔαν δύο τριγωνα τδ/υο πλευρ[τᾱιξ] δυσπλευρᾱιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα καὶ τγων/ιαν τ=ῆ γων/ιᾱ)/ισῃν)/εθῃ τ(υπτ=ων)/ισων ἐυθεῖων περιεθομ/ενην, καὶ τβ/ασιν τβ/ασει)/ισῃν (/εχει, καὶ ττρ/ιγωνον τ=ῶ τριγ/ωνῶ)/ισον)/εσται, καὶ αἱ λοιπαὶ γων/ιαι τᾱιξ λοιπᾱιξ γων/ιαιξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (αἱ)/ισαι πλευρᾱι (υποτέινουσιν.

)εστω δύο τριγωνα ταβγ, δεζ τδ/υο πλευρταβ, αγ τᾱιξ δυσπλευρᾱιξ τᾱιξ δε, δζ)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα τμαβ τ=ῆ δε τδαγ τ=ῆ δζ καὶ γων/ιαν τ(υπβαγ γων/ιᾱ τ=ῆ (υπεδζ)/ισῃν. λεγω, (/οτι καὶ β/ασιξ (ῆ βγ β/ασει τ=ῆ εζ)/ισῃ)εστ/ιν, καὶ ταβγ τριγωνον τ=ῶ δεζ τριγ/ωνῶ)/ισον)/εσται, καὶ αἱ λοιπαὶ γων/ιαι τᾱιξ λοιπᾱιξ γων/ιαιξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (αἱ)/ισαι πλευρᾱι (υποτέινουσιν, (ῆ μ(υπαβγ τ=ῆ (υπεδζ, (ῆ δ(υπαγβ τ=ῆ (υπεδζε.

)εφαρμοζομ/ενου γτο=υ αβγ τριγ/ωνου)επτδεζ τριγωνον καὶ τιθεμ/ενου το=υ μα σημείου)επτδ σημειον τ=ῆξ δαβ ἐυθείαξ)επτδε,)εφαρμ/οσει καὶ τβ σημειον)επτε διὰ τ)/ισῃν ἐ=ιναι ταβ τ=ῆ δε;)εφαρμος/ασῆξ δτ=ῆξ αβ)επτδε)εφαρμ/οσει καὶ (ῆ αγ ἐυθε=ια)επτδζ διὰ τ)/ισῃν ἐ=ιναι τ(υπβαγ γων/ιαν τ=ῆ (υπεδζ; (/ωστε καὶ τγ σημειον)επτζ σημειον)εφαρμ/οσει διὰ τ)/ισῃν π/αλιν ἐ=ιναι ταγ τ=ῆ δζ.)αλλμκαὶ τβ)επτε)εφηρμ/οκει; (/ωστε β/ασιξ (ῆ βγ)επβ/ασιν τεζ)εφαρμ/οσει. ἐι γτο=υ μβ)επτε)εφαρμ/οσαντοξ το=υ δγ)επτζ (ῆ βγ β/ασιξ)επτεζ οὐκ)εφαρμ/οσει, δ/υο ἐυθε=ιαι θωρ/ιον περιέχουσιν; (/οπερ)εστ)αδ/υνατον.)εφαρμ/οσει)/αρα (ῆ βγ β/ασιξ)επτεζ καὶ)/ισῃ α)υτ=ῆ)/εσται; (/ωστε καὶ (/ολον ταβγ τριγωνον)επ(/ολον τδεζ τριγωνον)εφαρμ/οσει καὶ)/ισον α)υτ=ῶ)/εσται, καὶ αἱ λοιπαὶ γων/ιαι)επτλοιπγων/ιαξ)εφαρμ/οσουσι καὶ)/ισαι α)υτᾱιξ)/εσονται, (ῆ μ(υπαβγ τ=ῆ (υπεδζ (ῆ δ(υπαγβ τ=ῆ (υπεδζε.

ἔαν)/αρα δύο τριγωνα τδ/υο πλευρ[τᾱιξ] δ/υο πλευρᾱιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα καὶ τγων/ιαν τ=ῆ γων/ιᾱ)/ισῃν)/εθῃ τ(υπτ=ων)/ισων ἐυθεῖων περιεθομ/ενην, καὶ τβ/ασιν τβ/ασει)/ισῃν (/εχει, καὶ ττρ/ιγωνον τ=ῶ τριγ/ωνῶ)/ισον)/εσται, καὶ αἱ λοιπαὶ γων/ιαι τᾱιξ λοιπᾱιξ γων/ιαιξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (αἱ)/ισαι πλευρᾱι (υποτέινουσιν; (/οπερ)/εδει δε=ιχαί.

Proposition 4

If two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . And (let) the angle BAC (be) equal to the angle EDF . I say that the base BC is also equal to the base EF , and triangle ABC will be equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (That is) ABC to DEF , and ACB to DFE .

For if triangle ABC is applied to triangle DEF ,[†] the point A being placed on the point D , and the straight-line AB on DE , (then) the point B will also coincide with E , on account of AB being equal to DE . So (because of) AB coinciding with DE , the straight-line AC will also coincide with DF , on account of the angle BAC being equal to EDF . So the point C will also coincide with the point F , again on account of AC being equal to DF . But, point B certainly also coincided with point E , so that the base BC will coincide with the base EF . For if B coincides with E , and C with F , and the base BC does not coincide with EF , (then) two straight-lines will encompass an area. The very thing is impossible [Post. 1].[‡] Thus, the base BC will coincide with EF , and will be equal to it [C.N. 4]. So the whole triangle ABC will coincide with the whole triangle DEF , and will be equal to it [C.N. 4]. And the remaining angles will coincide with the remaining angles, and will be equal to them [C.N. 4]. (That is) ABC to DEF , and ACB to DFE [C.N. 4].

Thus, if two triangles have two sides equal to two sides, respectively, and have the angle(s) enclosed by the equal straight-lines equal, (then) they will also have the base equal to the base, and the triangle will be equal to the triangle, and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles. (Which is) the very thing it was required to show. [†] The application of one figure to another should be counted as an additional postulate.

[‡] Since Post. 1 implicitly assumes that the straight-line joining two given points is unique.

5.

τ=ων)ισοσκελ=ων τριγ/ωνων ἂι πρτ=η β/ασει γων/ιαι)/ισαι)αλλ/ηλαιξ εἰς/ιν, καὶ προσεκβληθεῖς=ων τ=ων)/ισων ἐυθεῖων ἂι (υπτβ/ασιν γων/ιαι)/ισαι)αλλ/ηλαιξ)/εσονται.

) /εστω τρ/ιγωνον)ισοσκελταβγ)/ισην)/εθον ταβ πλευρτ=η αγ πλευρ=α, καὶ προσεκβεβλ/ησθωσαν)επ) ἐυθείαξ τᾱιξ αβ, αγ ἐυθε=ιαι ἂι βδ, γε, λ/εγω, (/οτι (η μ(υπαβγ γων/ια τ=η (υπαγβ)/ιση)εστ/ιν, (η δ(υπαγβδ τ=η (υπβγε.

ἐλ/ηφθω γ)επτ=ηξ βδ τυθσημε=ιον τζ, καὶ)αφηρ/ησθω)απτ=ηξ μείζονοξ τ=ηξ αε τ=η)ελ/ασσονι τ=η αζ)/ιση (η αη, καὶ)επεζέυθθωσαν ἂι ζγ, ηβ ἐυθε=ιαι.

)επὲι ὁ=υν)/ιση)εστ(η μαζ τ=η αη (η δαβ τ=η αγ, δ/υο δᾱι ζα, αγ δυστᾱιξ ηα, αβ)/ισαι εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ιαν κοινπερίεθουσι τ(υπζαη; β/ασιξ)/αρα (η ζγ β/ασει τ=η ηβ)/ιση)εστ/ιν, καὶ ταζγ τρ/ιγωνον τ=ω αηβ τριγ/ωνω)/ισον)/εσται, καὶ ἂι λοιπᾱι γων/ιαι τᾱιξ λοιπᾱιξ γων/ιαιξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (ἂι)/ισαι πλευρᾱι (υποτέινουσιν, (η μ(υπαγζ τ=η (υπαβη, (η δ(υπαζγ τ=η (υπαηβ. καὶ)επὲι (/ολη (η αζ (/ολη τ=η αη)εστιν)/ιση, (=ων (η αβ τ=η αγ)εστιν)/ιση, λοιπ)/αρα (η βζ λοιπ=η τ=η γη)εστιν)/ιση.)εδείθθη δκαὶ (η ζγ τ=η ηβ)/ιση; δ/υο δᾱι βζ, ζγ δυστᾱιξ

γη, ηβ)/ισαι εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ια (η (υπβζγ γων/ια τη (υπηγβ)/ιση, καὶ β/ασιξ α)υτ=ων κοιν(η βγ; καὶ τβζγ)/αρα τρ/ιγωνων τ=ω γηβ τριγ/ωνω)/ισον)/εσται, καὶ αἱ λοιπαὶ γων/ιαὶ τᾶιξ λοιπᾶιξ γων/ιαῖξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (αἱ)/ισαι πλευρᾶι (υποτέινουσιν;)/ιση)/αρα)εστ(η μ(υπζβγ τ=η (υπηγβ (η δ(υπβγζ τ=η (υπηγβ.)επὲι ὁ=υν (/ολη (η (υπαβη γων/ια (/ολη τ=η (υπαγζ γων/ια)εδείθθη)/ιση, (=ων (η (υπηγβη τ=η (υπβγζ)/ιση, λοιπ)/αρα (η (υπαβγ λοιπ=η τ=η (υπαγβ)εστιν)/ιση; καὶ εἰσι πρτ=η β/ασει το=υ αβγ τριγ/ωνου.)εδείθθη δκαὶ (η (υπζβγ τ=η (υπηγβ)/ιση; καὶ εἰσιν (υπτβ/ασιν.

τ=ων)/αρα)ισοσκελ=ων τριγ/ωνων αἱ τρτ=η β/ασει γων/ια)/ισαι)αλλ/ηλαιξ εἰς/ιν, καὶ προσεκβληθεῖς=ων τ=ων)/ισων ἐυθεῖων αἱ (υπτβ/ασιν γων/ια)/ισαι)αλλ/ηλαιξ)/εσονται; (/οπερ)/εδει δε=ιχαί.

Proposition 5

For isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another.

Let ABC be an isosceles triangle having the side AB equal to the side AC , and let the straight-lines BD and CE be produced in a straight-line with AB and AC (respectively) [Post. 2]. I say that the angle ABC is equal to ACB , and (angle) CBD to BCE .

For let the point F be taken at random on BD , and let AG be cut off from the greater AE , equal to the lesser AF [Prop. 1.3]. Also, let the straight-lines FC and GB be joined [Post. 1].

In fact, since AF is equal to AG , and AB to AC , the two (straight-lines) FA , AC are equal to the two (straight-lines) GA , AB , respectively. They also encompass a common angle, FAG . Thus, the base FC is equal to the base GB , and the triangle AFC will be equal to the triangle AGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. (That is) ACF to ABG , and AFC to AGB . And since the whole of AF is equal to the whole of AG , within which AB is equal to AC , the remainder BF is thus equal to the remainder CG [C.N. 3]. But FC was also shown (to be) equal to GB . So the two (straight-lines) BF , FC are equal to the two (straight-lines) CG , GB , respectively, and the angle BFC (is) equal to the angle CGB , and the base BC is common to them. Thus, the triangle BFC will be equal to the triangle CGB , and the remaining angles subtended by the equal sides will be equal to the corresponding remaining angles [Prop. 1.4]. Thus, FBC is equal to GCB , and BCF to CBG . Therefore, since the whole angle ABG was shown (to be) equal to the whole angle ACF , within which CBG is equal to BCF , the remainder ABC is thus equal to the remainder ACB [C.N. 3]. And they are at the base of triangle ABC . And FBC was also shown (to be) equal to GCB . And they are under the base.

Thus, for isosceles triangles, the angles at the base are equal to one another, and if the equal sides are produced (then) the angles under the base will be equal to one another. (Which is) the very thing it was required to show.

6.

)ἐαν τριγ/ωνου αἱ δ/υο γων/ιαὶ)/ισαι)αλλ/ηλαιξ)=ωσιν, καὶ αἱ (υπτ)/ισαξ γων/ιαῖξ (υποτέινουσαι πλευρᾶι)/ισαι)αλλ/ηλαιξ)/εσονται.

)/εστω τρ/ιγωνον ταβγ)/ισην)/εθον τ(υπαβγ γων/ιαν τ=η (υπαγβ γων/ια; λεγω, (/οτι καὶ πλευρ(η αβ πλευρ=α τ=η αγ)εστιν)/ιση.

ἐἴ γ' /ανιζ/οξ)εστιν (ἡ ἀβ τ=ἡ αγ, (ἡ (ετ/ερα α)υτ=ων μείζων)εστ/ιν.)/εστω μείζων (ἡ ἀβ, καὶ)αφηρ/ησθω)απτ=ἡξ μείζονοξ τ=ἡξ ἀβ τ=ἡ)ελ/αττονι τ=ἡ αγ)/ιση (ἡ δβ, καὶ)επεξέυθθω (ἡ δγ.

)επεί ὁ=υν)/ιση)εστ(ἡ δβ τ=ἡ αγ κοινδ(ἡ βγ, δ/υο δᾱι δβ, βγ δ/υο τᾱιξ αγ, γβ)/ισαι εἰς(εκατ/ερα (εκατ/ερα, καὶ γων/ια (ἡ (υπδβγ γων/ια τ=ἡ (υπαγβ)εστιν)/ιση; β/ασιξ)/αρα (ἡ δγ β/ασει τ=ἡ ἀβ)/ιση)εστ/ιν, καὶ τδβγ τρ/ιγωνον τ=ω αγβ τριγ/ωνω)/ισον)/εσται, τ)/ελασσον τ=ω μείζονι; (/οπερ)/ατοπον; οὐκ)/αρα)/ανιζ/οξ)εστιν (ἡ ἀβ τ=ἡ αγ;)/ιση)/αρα.

)ἐὰν)/αρα τριγ/ωνου ᾱι δ(υο γων/ιαι)/ισαι)αλλ/ηλαιοξ)=ωσιν, καὶ ᾱι (υπτ)/ισαξ γων/ιαξ (υποτέινουσαι πλευρᾱι)/ισαι)αλλ/ηλαιοξ)/εσονται; (/οπερ)/εδει δε=ιχαί.

Proposition 6

If a triangle has two angles equal to one another (then) the sides subtending the equal angles will also be equal to one another.

Let ABC be a triangle having the angle ABC equal to the angle ACB . I say that side AB is also equal to side AC .

For if AB is unequal to AC (then) one of them is greater. Let AB be greater. And let DB , equal to the lesser AC , be cut off from the greater AB [Prop. 1.3]. And let DC be joined [Post. 1].

Therefore, since DB is equal to AC , and BC (is) common, the two sides DB , BC are equal to the two sides AC , CB , respectively, and the angle DBC is equal to the angle ACB . Thus, the base DC is equal to the base AB , and the triangle DBC will be equal to the triangle ACB [Prop. 1.4], the lesser to the greater. The very notion (is) absurd [C.N. 5]. Thus, AB is not unequal to AC . Thus, (it is) equal.[†]

Thus, if a triangle has two angles equal to one another (then) the sides subtending the equal angles will also be equal to one another. (Which is) the very thing it was required to show. [†] Here, use is made of the previously unmentioned common notion that if two quantities are not unequal then they must be equal. Later on, use is made of the closely related common notion that if two quantities are not greater than or less than one another, respectively, then they must be equal to one another.

7.

)επτ=ἡξ α)υτ=ἡξ ἐυθείαξ δ/υο τᾱιξ α)υτᾱιξ ἐυθείαιοξ)/αλλαι δ/υο ἐυθε=ιαι)/ισαι (εκατ/ερα (εκατ/ερα ὅν συσταθ/ησονται πρ)/αλλω καὶ)/αλλω σημείω)επτα)υτμ/ερη τα)υτπ/ερατα)/εθουσαι τᾱιξ)εχ)αρθ=ἡξ ἐυθείαιοξ.

ἐἴ γδυνατ/ον,)επτ=ἡξ α)υτ=ἡξ ἐυθείαξ τ=ἡξ ἀβ δ/υο τᾱιξ α)υτᾱιξ ἐυθείαιοξ τᾱιξ αγ, γβ)/αλλαι δ/υο ἐυθε=ιαι ᾱι αδ, δβ)/ισαι (εκατ/ερα (εκατ/ερα συνεστ/ατωσαν πρ)/αλλω καὶ)/αλλω σημείω τ=ω τε γ καὶ δ)επτα)υτμ/ερη τα)υτπ/ερατα)/εθουσαι, (/ωστε)/ισην ἐ=ιναι τμγα τ=ἡ δα τα)υτπ/εραξ)/εθουσαν α)υτ=ἡ τα, τδγβ τ=ἡ δβ τα)υτπ/εραξ)/εθουσαν α)υτ=ἡ ββ, καὶ)επεξέυθθω (ἡ γδ.

)επεί ὁ=υν)/ιση)εστ(ἡ αγ τ=ἡ αδ,)/ιση)εστκαὶ γων/ια (ἡ (υπαγδ τ=ἡ (υπαδγ; μείζων)/αρα (ἡ (υπαδγ τ=ἡξ (υπδγβ; πολλ=ω)/αρα (ἡ (υπγδβ μείζων)εστ/ι τ=ἡξ (υπδγβ.

π/άλιν)επεί)/ιση)εστ(η γβ τ=η δβ,)/ιση)εστκαὶ γων/ια (η (υπγδβ γων/ια τ=η (υπδγβ.)εδείθθη δα)υτ=ηξ καὶ πολλ=ω μείζων; (/οπερ)εστ)αδ/υνατον.

ὅυκ)/αρα)επτ=ηξ α)υτ=ηξ ἐυθείαξ δ/υο τᾶιξ α)υτᾶιξ ἐυθείαιξ)/αλλαι δ/υο ἐυθε=ιαι)/ισαι (εκατ/ερα (εκατ/ερα συσταθ/ησονται πρ)/αλλω καὶ)/αλλω σημείω)επτα)υτμ/ερη τα)υτπ/ερατα)/εθουσαι τᾶιξ)εχ)αρθ=ηξ ἐυθείαιξ; (/οπερ)εδει δε=ιχαί.

Proposition 7

On the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines.

For, if possible, let the two straight-lines AC , CB , equal to two other straight-lines AD , DB , respectively, be constructed on the same straight-line AB , meeting at different points, C and D , on the same side (of AB), and having the same ends (on AB). So CA is equal to DA , having the same end A as it, and CB is equal to DB , having the same end B as it. And let CD be joined [Post. 1].

Therefore, since AC is equal to AD , the angle ACD is also equal to angle ADC [Prop. 1.5]. Thus, ADC (is) greater than DCB [C.N. 5]. Thus, CDB is much greater than DCB [C.N. 5]. Again, since CB is equal to DB , the angle CDB is also equal to angle DCB [Prop. 1.5]. But it was shown that the former (angle) is also much greater (than the latter). The very thing is impossible.

Thus, on the same straight-line, two other straight-lines equal, respectively, to two (given) straight-lines (which meet) cannot be constructed (meeting) at a different point on the same side (of the straight-line), but having the same ends as the given straight-lines. (Which is) the very thing it was required to show.

8.

)ἐὰν δ/υο τρ/ιγωνα τδ/υο πλευρ[τᾶιξ] δ/υο πλευρᾶιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα,)/εθῃ δκαὶ τβ/ασιν τ=η β/ασει)/ισην, καὶ τγων/ιαν τ=η γων/ια)/ισην (/εχει τ(υπτ=ων)/ισων ἐυθεῖων περιεθομ/ενην.

)εστω δ/υο τρ/ιγωνα ταβγ, δεζ τδ/υο πλευρταβ, αγ τᾶιξ δ/υο πλευρᾶιξ τᾶιξ δε, δζ)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα, τμαβ τ=η δε τδαγ τ=η δζ;)εθ/ετω δκαὶ β/ασιν τβγ β/ασει τ=η εζ)/ισην; λ/εγω, (/οτι καὶ γων/ια (η (υπβαγ γων/ια τ=η (υπεδζ)εστιν)/ιση.

)εφαρμοζομ/ενου γτο=υ αβγ τριγ/ωνου)επτδεζ τρ/ιγωνον καὶ τιθεμ/ενου το=υ μβ σημείου)επτε σημ=ιον τ=ηξ δβγ ἐυθείαξ)επτεζ)εφαρμ/οσει καὶ τγ σημ=ιον)επτζ δια τ)/ισην ἐ=ιναι τβγ τ=η εζ;)εφαρμοζ/ασηξ δτ=ηξ βγ)επτεζ)εφαρμ/οσουσι καὶ αὶ βα, γα)επτεδ, δζ. εἰ γβ/ασιξ μ(η βγ)επβ/ασιν τεζ)εφαρμ/οσει, αὶ δβα, αγ πλευρὰι)επτεδ, δζ ὅυκ)εφαρμ/οσουσιν)αλλπαρὰλλ/αχουσιν (ωξ αὶ εη, ηζ, συσταθ/ησονται)επτ=ηξ α)υτ=ηξ ἐυθείαξ δ/υο τᾶιξ α)υτᾶιξ ἐυθείαιξ)/αλλαι δ/υο ἐυθε=ιαι)/ισαι (εκατ/ερα (εκατ/ερα πρ)/αλλω καὶ)/αλλω σημείω)επτα)υτμ/ερη τα)υτπ/ερατα)/εθουσαι. ὅυ συν/ιστανται δ/ε; ὅυκ)/αρα)εφαρμοζομ/ενηξ τ=ηξ βγ β/ασεωξ)επτεζ β/ασιν ὅυκ)εφαρμ/οσουσι καὶ αὶ βα, αγ πλευρὰι)επτεδ, δζ.)εφαρμ/οσουσιν)/αρα; (/ωστε καὶ γων/ια (η (υπβαγ)επγων/ιαν τ(υπεδζ)εφαρμ/οσει καὶ)/ιση α)υτ=η)/εσταί.

)ἐὰν)/αρα δ/υο τρ/ιγωνα τδ/υο πλευρ[τᾶιξ] δ/υο πλευρᾶιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα καὶ τβ/ασιν τ=η β/ασει)/ισην)/εθῃ, καὶ τγων/ιαν τ=η γων/ια)/ισην (/εχει τ(υπτ=ων)/ισων ἐυθεῖων περιεθομ/ενην; (/οπερ)εδει δε=ιχαί.

Proposition 8

If two triangles have two sides equal to two sides, respectively, and also have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines.

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is) AB to DE , and AC to DF . Let them also have the base BC equal to the base EF . I say that the angle BAC is also equal to the angle EDF .

For if triangle ABC is applied to triangle DEF , the point B being placed on point E , and the straight-line BC on EF , (then) point C will also coincide with F , on account of BC being equal to EF . So (because of) BC coinciding with EF , (the sides) BA and CA will also coincide with ED and DF (respectively). For if base BC coincides with base EF , but the sides AB and AC do not coincide with ED and DF (respectively), but miss like EG and GF (in the above figure), then we will have constructed upon the same straight-line two other straight-lines equal, respectively, to two (given) straight-lines, and (meeting) at a different point on the same side (of the straight-line), but having the same ends. But (such straight-lines) cannot be constructed [Prop. 1.7]. Thus, the base BC being applied to the base EF , the sides BA and AC cannot not coincide with ED and DF (respectively). Thus, they will coincide. So the angle BAC will also coincide with angle EDF , and will be equal to it [C.N. 4].

Thus, if two triangles have two sides equal to two side, respectively, and have the base equal to the base, (then) they will also have equal the angles encompassed by the equal straight-lines. (Which is) the very thing it was required to show.

9.

τιδοθε=ισαν γων/ιαν εϑ/υγραμμον διθα τεμε=ιν.

)εστω (η δοθε=ισα γων/ια εϑ/υγραμμοξ (η (υπβαγ. δε=ι δα)υτδιθα τεμε=ιν.

ειληφθω)επτ=ηξ αβ τυθσημε=ιον τδ, και)αφηρ/ησθω)απτ=ηξ αγ τ=η αδ)/ιση (η αε, και)επεξευθθω (η δε, και συνεστ/ατω)επτ=ηξ δε τρ/ιγωνον)ις/οπλευρον τδεξ, και)επεξευθθω (η αζ; λεγω, (/οτι (η (υπβαγ γων/ια διθα τ/ετμηται (υπτ=ηξ αζ ευθειαξ.

)επει γ)/ιση)εστ(η αδ τ=η αε, κοινδ(η αζ, δι/υο δαι δα, αζ δυσταιξ εα, αζ)/ισαι εις(εκατ/ερα (εκατ/ερα. και β/ασιξ (η δζ β/ασει τ=η εζ)/ιση)εστ/ιν; γων/ια)/αρα (η (υπδαζ γων/ια τ=η (υπεαζ)/ιση)εστ/ιν.

(η)/αρα δοθε=ισα γων/ια εϑ/υγραμμοξ (η (υπβαγ διθα τ/ετμηται (υπτ=ηξ αζ ευθειαξ; (/οπερ)εδει ποιησαι.

Proposition 9

To cut a given rectilinear angle in half.

Let BAC be the given rectilinear angle. So it is required to cut it in half.

Let the point D be taken at random on AB , and let AE , equal to AD , be cut off from AC [Prop. 1.3], and let DE be joined. And let the equilateral triangle DEF be constructed upon DE [Prop. 1.1], and let AF be joined. I say that the angle BAC has been cut in half by the straight-line AF .

For since AD is equal to AE , and AF is common, the two (straight-lines) DA , AF are equal to the two (straight-lines) EA , AF , respectively. And the base DF is equal to the base EF . Thus, angle DAF is equal to angle EAF [Prop. 1.8].

Thus, the given rectilinear angle BAC has been cut in half by the straight-line AF . (Which is) the very thing it was required to do.

10.

τδοθε=ισαν ἐυθε=ιαν πεπερασμ/ενην δι/ιθα τεμε=ιν.
)/εστω (η δοθε=ισα ἐυθε=ια πεπερασμ/ενη (η αβ; δε=ι δταβ ἐυθε=ιαν πεπερασμ/ενην δι/ιθα τεμε=ιν.

συνεστ/ατω)επ) α)υτ=ηξ τρ/ιγωνον)ις/οπλευρον ταβγ, καὶ τετμ/ησθω (η (υπαγβ γων/ια δι/ιθα τ=η γδ ἐυθεία; λ/εγω, (/οτι (η αβ ἐυθε=ια δι/ιθα τ/ετμηται καττδ σημει=ιον.

)επὲι γ)/ιση)εστ(η αγ τ=η γβ, κοινδ(η γδ, δ/υο δ'αί αγ, γδ δ/υο τᾶιξ βγ, γδ)/ισαί εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ια (η (υπαγδ γων/ια τ=η (υπβγδ)/ιση)εστ/ιν; β/ασιξ)/αρα (η αδ β/ασει τ=η βδ)/ιση)εστ/ιν.

(η)/αρα δοθε=ισα ἐυθε=ια πεπερασμ/ενη (η αβ δι/ιθα τ/ετμηται καττδ; (/οπερ)/εδει ποιῆσαι.

Proposition 10

To cut a given finite straight-line in half.

Let AB be the given finite straight-line. So it is required to cut the finite straight-line AB in half.

Let the equilateral triangle ABC be constructed upon (AB) [Prop. 1.1], and let the angle ACB be cut in half by the straight-line CD [Prop. 1.9]. I say that the straight-line AB has been cut in half at point D .

For since AC is equal to CB , and CD (is) common, the two (straight-lines) AC, CD are equal to the two (straight-lines) BC, CD , respectively. And the angle ACD is equal to the angle BCD . Thus, the base AD is equal to the base BD [Prop. 1.4].

Thus, the given finite straight-line AB has been cut in half at (point) D . (Which is) the very thing it was required to do.

11.

τ=η δοθείση ἐυθεία)απτο=υ πρα)υτ=η δοθ/εντοξ σημείου πρ)ορθγων/ιαξ ἐυθε=ιαν γραμμ)αγαγε=ιν.

)εστω (η μδοθε=ισα ἐυθε=ια (η αβ τδοθoσημε=ιον)επ) α)υτ=ηξ τγ; δε=ι δ)απτο=υ γ σημείου τ=η αβ ἐυθεία πρ)ορθγων/ιαξ ἐυθε=ιαν γραμμ)αγαγε=ιν.

ἐλ/ηφθω)επτ=ηξ αγ τυθσημε=ιον τδ, καὶ κείσθω τ=η γδ)/ιση (η γε, καὶ συνεστ/ατω)επτ=ηξ δε τρ/ιγωνον)ις/οπλευρον τζδε, καὶ)επεξέυθθω (η ζγ; λ/εγω, (/οτι τ=η δοθείση ἐυθεία τ=η αβ)απτο=υ πρα)υτ=η δοθ/εντοξ σημείου το=υ γ πρ)ορθγων/ιαξ ἐυθε=ια γραμμ)=ηκται (η ζγ.

)επὲι γ)/ιση)εστ(η δγ τ=η γε, κοινδ(η γζ, δ/υο δάι δγ, γζ δυστᾶιξ εγ, γζ)/ισαι εἰς(εκατ/ερα (εκατ/ερα; καὶ β/ασιξ (η δζ β/ασει τ=η ζε)/ιση)εστ/ιν; γων/ια)/αρα (η (υπδγζ γων/ια τ=η (υπεγζ)/ιση)εστ/ιν; καὶ εἰσιν)εφεχ=ηξ. (/οταν δέυθε=ια)επ) ἐυθε=ιαν σταθε=ισα τ)εφεχ=ηξ γων/ιαξ)/ισαξ)αλλ/ηλαιξ ποῖη,)ορθ(εκατ/ερα τ=ων)/ισων γωνῶν)εστιν;)ορθ)/αρα)εστ(εκατ/ερα τ=ων (υπδγζ, ζγε.

τ=η)/αρα δοθείση ἐυθεία τ=η αβ)απτο=υ πρα)υτ=η δοθ/εντοξ σημείου το=υ γ πρ)ορθγων/ιαξ ἐυθε=ια γραμμ)=ηκται (η γζ; (/οπερ)/εδει ποῖησαι.

Proposition 11

To draw a straight-line at right-angles to a given straight-line from a given point on it.

Let AB be the given straight-line, and C the given point on it. So it is required to draw a straight-line from the point C at right-angles to the straight-line AB .

Let the point D be taken at random on AC , and let CE be made equal to CD [Prop. 1.3], and let the equilateral triangle FDE be constructed on DE [Prop. 1.1], and let FC be joined. I say that the straight-line FC has been drawn at right-angles to the given straight-line AB from the given point C on it.

For since DC is equal to CE , and CF is common, the two (straight-lines) DC , CF are equal to the two (straight-lines), EC , CF , respectively. And the base DF is equal to the base FE . Thus, the angle DCF is equal to the angle ECF [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle [Def. 1.10]. Thus, each of the (angles) DCF and FCE is a right-angle.

Thus, the straight-line CF has been drawn at right-angles to the given straight-line AB from the given point C on it. (Which is) the very thing it was required to do.

12.

)επτδοθε=ισαν ἐυθε=ιαν)/απειρον)απτο=υ δοθ/εντοξ σημείου, (μ/η)εστιν)επ) α)υτ=ηξ, κ/αθετον ἐυθε=ιαν γραμμ)αγαγε=ιν.

)εστω (η μδοθε=ισα ἐυθε=ια)/απειροξ (η αβ τδδοθσημε=ιον, (μ/η)εστιν)επ) α)υτ=ηξ, τγ; δε=ι δ)επτδοθε=ισαν ἐυθε=ιαν)/απειρον ταβ)απτο=υ δοθ/εντοξ σημείου το=υ γ, (μ/η)εστιν)επ) α)υτ=ηξ, κ/αθετον ἐυθε=ιαν γραμμ)αγαγε=ιν.

ἐλ/ηφθω γ)επ(ετερα μ/ερη τ=ηξ αβ ἐυθείαξ τυθσημε=ιον τδ, καὶ κ/εντρῷ μτ=ω γ διαστ/ηματι δτ=ω γδ κ/υκλοξ γεγρ/αφθω (ο εζη, καὶ τετμ/ησθω (η εη ἐυθε=ια δ/ιθα κατθ, καὶ)επεξέυθθωσαν αἱ γη, γθ, γε ἐυθε=ιαι; λ/εγω, (/οτι)επτδοθε=ισαν ἐυθε=ιαν)/απειρον ταβ)απτο=υ δοθ/εντοξ σημείου το=υ γ, (μ/η)εστιν)επ) α)υτ=ηξ, κ/αθετοξ)=ηκται (η γθ.

)επὲι γ)/ιση)εστ(η ηθ τ=η θε, κοινδ(η θγ, δ/υο δάι ηθ, θγ δ/υο τᾶιξ εθ, θγ)/ισαι εἰς(εκατ/ερα (εκατ/ερα; καὶ β/ασιξ (η γη β/ασει τ=η γε)εστιν)/ιση; γων/ια)/αρα (η (υπγθη γων/ια τ=η (υπεθγ)εστιν)/ιση. καὶ εἰσιν)εφεχ=ηξ. (/οταν δέυθε=ια)επ) ἐυθε=ιαν

σταθε=ισα τ)εφεχ=ηξ γων/ιαξ)/ισαξ)αλλ/ηλαιξ ποῖη,)ορθ(εκατ/ερα τ=ων)/ισων γωνῶν)εστιν, καὶ (η)εφεστηκῶια εὐθε=ια κ/αθετοξ καλε=ιται)εφ))εφ/εστηκεν.

)επιδοδοθε=ισαν)/αρα εὐθε=ιαν)/απειρον ταβ)απτο=υ δοθ/εντοξ σημείου το=υ γ, (μ/η)εστιν)επ) α)υτ=ηξ, κ/αθετοξ =)ηκται (η γθ; (/οπερ)/εδει ποῖησαι.

Proposition 12

To draw a straight-line perpendicular to a given infinite straight-line from a given point which is not on it.

Let AB be the given infinite straight-line and C the given point, which is not on it. So it is required to draw a straight-line perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For let point D be taken at random on the other side (to C) of the straight-line AB , and let the circle EFG be drawn with center C and radius CD [Post. 3], and let the straight-line EG be cut in half at (point) H [Prop. 1.10], and let the straight-lines CG , CH , and CE be joined. I say that the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it.

For since GH is equal to HE , and HC (is) common, the two (straight-lines) GH , HC are equal to the two (straight-lines) EH , HC , respectively, and the base CG is equal to the base CE . Thus, the angle CHG is equal to the angle EHG [Prop. 1.8], and they are adjacent. But when a straight-line stood on a(nother) straight-line makes the adjacent angles equal to one another, each of the equal angles is a right-angle, and the former straight-line is called a perpendicular to that upon which it stands [Def. 1.10].

Thus, the (straight-line) CH has been drawn perpendicular to the given infinite straight-line AB from the given point C , which is not on it. (Which is) the very thing it was required to do.

13.

)εαν εὐθε=ια)επ) εὐθε=ιαν σταθε=ισα γων/ιαξ ποῖη,)/ητοι δ/υο)ορθ(δυσ)ορθᾶιξ)/ισαξ ποῖησει.

εὐθε=ια γ/αρ τιξ (η αβ))επ) εὐθε=ιαν τγδ σταθε=ισα γων/ιαξ ποιείτω τ(υπγβα, αβδ; λ, (/οτι αἱ (υπγβα, αβδ γων/ιαί)/ητοι δ/υο)ορθάι εἰσιν)δυσ)ορθᾶιξ)/ισαί.

εἰ μὲ=υν)/ιση)εστ(η (υπγβα τ=η (υπαβδ, δ/υο)ορθάι εἰσιν. εἰ δ'οὐ,)/ηθθω)απτο=υ β σημείου τ=η γδ [εὐθεία] πρ)ορθ(η βε; αἱ)/αρα (υπγβε, εβδ δ/υο)ορθάι εἰσιν; καὶ)επὲι (η (υπγβε δυστᾶιξ (υπγβα, αβε))/ιση)εστ/ιν, κοινπροσκέισθω (η (υπεβδ; αἱ)/αρα (υπγβε, εβδ τριςτᾶιξ (υπγβα, αβε, εβδ))/ισαί εἰς/ιν. π/αλιν,)επὲι (η (υπδβα δυστᾶιξ (υπδβε, εβα))/ιση)εστ/ιν, κοινπροσκέισθω (η (υπαβγ; αἱ)/αρα (υπδβα, αβγ τριςτᾶιξ (υπδβε, εβα, αβγ))/ισαί εἰς/ιν.)εδείθθησαν δκαὶ αἱ (υπγβε, εβδ τριςτᾶιξ α)υτᾶιξ)/ισαί; τδτ=ω α)υτ=ω)/ισα καὶ)αλλ/ηλοιξ)εστ)/ισα; καὶ αἱ (υπγβε, εβδ))/αρα τᾶιξ (υπδβα, αβγ))/ισαί εἰς/ιν;)αλλάι (υπγβε, εβδ δ/υο)ορθάι εἰσιν; καὶ αἱ (υπδβα, αβγ))/αρα δυσ)ορθᾶιξ)/ισαί εἰς/ιν.

)εαν)/αρα εὐθε=ια)επ) εὐθε=ιαν σταθε=ισα γων/ιαξ ποῖη,)/ητοι δ/υο)ορθ(δυσ)ορθᾶιξ)/ισαξ ποῖησει; (/οπερ)/εδει δε=ιχαί.

Proposition 13

If a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles.

For let some straight-line AB stood on the straight-line CD make the angles CBA and ABD . I say that the angles CBA and ABD are certainly either two right-angles, or (have a sum) equal to two right-angles.

In fact, if CBA is equal to ABD (then) they are two right-angles [Def. 1.10]. But, if not, let BE be drawn from the point B at right-angles to [the straight-line] CD [Prop. 1.11]. Thus, CBE and EBD are two right-angles. And since CBE is equal to the two (angles) CBA and ABE , let EBD be added to both. Thus, the (sum of the angles) CBE and EBD is equal to the (sum of the) three (angles) CBA , ABE , and EBD [C.N. 2]. Again, since DBA is equal to the two (angles) DBE and EBA , let ABC be added to both. Thus, the (sum of the angles) DBA and ABC is equal to the (sum of the) three (angles) DBE , EBA , and ABC [C.N. 2]. But (the sum of) CBE and EBD was also shown (to be) equal to the (sum of the) same three (angles). And things equal to the same thing are also equal to one another [C.N. 1]. Therefore, (the sum of) CBE and EBD is also equal to (the sum of) DBA and ABC . But, (the sum of) CBE and EBD is two right-angles. Thus, (the sum of) ABD and ABC is also equal to two right-angles.

Thus, if a straight-line stood on a(nother) straight-line makes angles, it will certainly either make two right-angles, or (angles whose sum is) equal to two right-angles. (Which is) the very thing it was required to show.

14.

ἔαν πρ/οξ τινι ἐυθείᾳ καὶ τ=ω πρ(α)υτ=η σημείω δ/υο ἐυθε=ιαὶ μ(ε)π(τ)α)υτμ/ερη κείμεναι τ)εφεχ=ηξ γων/ιαξ δυς)ορθᾷξ)/ισαξ ποῖωσιν,)επ) ἐυθείαξ)/εσονται)αλλ/ηλαιξ αὶ ἐυθε=ιαὶ.

πργ/αρ τινι ἐυθείᾳ τ=η αβ καὶ τ=ω πρ(α)υτ=η σημείω τ=ω β δ/υο ἐυθε=ιαὶ αὶ βγ, βδ μ(ε)π(τ)α)υτμ/ερη κείμεναι τ)εφεχ=ηξ γων/ιαξ τ(υπαβγ, αβδ δ/υο)ορθᾷξ)/ισαξ ποιείτωσαν; λ/εγω, (/οτι))επ) ἐυθείαξ)εσττ=η γβ (η βδ).

εἰ γμ/η)εστι τ=η βγ)επ) ἐυθείαξ (η βδ,)/εστω τ=η γβ)επ) ἐυθείαξ (η βε.

)επὲι ὁ=υν ἐυθε=ια (η αβ))επ) ἐυθε=ιαν τγβε)εφ/εστηκεν, αὶ)/αρα (υπαβγ, αβε γων/ιαὶ δ/υο)ορθᾷξ)/ισαὶ εἰς/ιν; εἰςδκαὶ αὶ (υπαβγ, αβδ δ/υο)ορθᾷξ)/ισαὶ; αὶ)/αρα (υπγβα, αβε τᾷξ (υπγβα, αβδ)/ισαὶ εἰς/ιν. κοιν)αφηρ/ησθω (η (υπγβα; λοιπ)/αρα (η (υπαβε λοιπ=η τ=η (υπαβδ)εστιν)/ιση, (η)ελ/ασσων τ=η μείζονι; (/οπερ)εστ)αδ/υνατον. οὐκ)/αρα)επ) ἐυθείαξ)εστ(η βε τ=η γβ. (ομόιωξ δδείχομεν, (/οτι οὐδ)/αλλη τιξ πλτ=ηξ βδ;)επ) ἐυθείαξ)/αρα)εστ(η γβ τ=η βδ.

)ἔαν)/αρα πρ/οξ τινι ἐυθείᾳ καὶ τ=ω πρ(α)υτ=η σημείω δ/υο ἐυθε=ιαὶ μ(ε)π(α)υτμ/ερη κείμεναι τ)εφεχ=ηξ γων/ιαξ δυς)ορθᾷξ)/ισαξ ποῖωσιν,)επ) ἐυθείαξ)/εσονται)αλλ/ηλαιξ αὶ ἐυθε=ιαὶ; (/οπερ)εδει δε=ιχαὶ.

Proposition 14

If two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another.

For let two straight-lines BC and BD , not lying on the same side, make adjacent angles ABC and ABD (whose sum is) equal to two right-angles with some straight-line AB , at the point B on it. I say that BD is straight-on with respect to CB .

For if BD is not straight-on to BC (then) let BE be straight-on to CB .

Therefore, since the straight-line AB stands on the straight-line CBE , the (sum of the) angles ABC and ABE is thus equal to two right-angles [Prop. 1.13]. But (the sum of) ABC and ABD is also equal to two right-angles. Thus, (the sum of angles) CBA and ABE is equal to (the sum of angles) CBA and ABD [C.N. 1]. Let (angle) CBA be subtracted from both. Thus, the remainder ABE is equal to the remainder ABD [C.N. 3], the lesser to the greater. The very thing is impossible. Thus, BE is not straight-on with respect to CB . Similarly, we can show that neither (is) any other (straight-line) than BD . Thus, CB is straight-on with respect to BD .

Thus, if two straight-lines, not lying on the same side, make adjacent angles (whose sum is) equal to two right-angles with some straight-line, at a point on it, (then) the two straight-lines will be straight-on (with respect) to one another. (Which is) the very thing it was required to show.

15.

ἐὰν δ/υο ἐυθεῖαι τ/εμνωσιν ἀλλ/ηλαξ, τκατκορυφων/ιαξ)/ισαξ ἀλλ/ηλαιξ ποιο=υσιν.
 δ/υο γ'ἐυθεῖαι αἱ αβ, γδ τεμν/ετωσαν ἀλλ/ηλαξ καττε σημει=ον; λ/εγω, (/οτι)/ιση
)εστ(η μ(υπαεγ γων/ια τ=η (υπεδβ, (η δ(υπεγβ τ=η (υπαεδ.

ἔπει γ'ἐυθεῖαι (η αε) επ) ἐυθεῖαν τγδ)εφ/εστηκε γων/ιαξ ποιο=υσα τ(υπεγα, αεδ, αἱ)/αρα (υπεγα, αεδ γων/ιαι δυσ)ορθᾷξ)/ισαι εἰς/ιν. π/αλιν,)επεί ἐυθεῖαι (η δε) επ) ἐυθεῖαν ταβ)εφ/εστηκε γων/ιαξ ποιο=υσα τ(υπαεδ, δεβ, αἱ)/αρα (υπαεδ, δεβ γων/ιαι δυσ)ορθᾷξ)/ισαι εἰς/ιν.)εδείθησαν δκαὶ αἱ (υπεγα, αεδ δυσ)ορθᾷξ)/ισαι; αἱ)/αρα (υπεγα, αεδ τᾷξ (υπαεδ, δεβ)/ισαι εἰς/ιν. κοιν)αφηρ/ησθω (η (υπαεδ; λοιπ)/αρα (η (υπεγα λοιπ=η τ=η (υπεδ)/ιση)εστ/ιν; (ομώωξ δδειθ/ησεται, (/οτι καὶ αἱ (υπεγβ, δεα)/ισαι εἰς/ιν.

ἐὰν)/αρα δ/υο ἐυθεῖαι τ/εμνωσιν ἀλλ/ηλαξ, τκατκορυφων/ιαξ)/ισαξ ἀλλ/ηλαιξ ποιο=υσιν; (/οπερ)/εδει δε=ιχαί.

Proposition 15

If two straight-lines cut one another (then) they make the vertically opposite angles equal to one another.

For let the two straight-lines AB and CD cut one another at the point E . I say that angle AEC is equal to (angle) DEB , and (angle) CEB to (angle) AED .

For since the straight-line AE stands on the straight-line CD , making the angles CEA and AED , the (sum of the) angles CEA and AED is thus equal to two right-angles [Prop. 1.13]. Again, since the straight-line DE stands on the straight-line AB , making the angles AED and DEB , the (sum of the) angles AED and DEB is thus equal to two right-angles [Prop. 1.13]. But (the sum of) CEA and AED was also shown (to be) equal to two right-angles. Thus, (the sum of) CEA and AED is equal to (the sum of) AED and DEB [C.N. 1]. Let AED be subtracted from both. Thus, the remainder CEA is equal to the remainder DEB [C.N. 3]. Similarly, it can be shown that CEB and DEA are also equal.

Thus, if two straight-lines cut one another (then) they make the vertically opposite angles equal to one another. (Which is) the very thing it was required to show.

16.

παντριγωνου μιᾶς τῶν πλευρῶν προσεκβληθείσης (ἡ) ἐκτῶν/ια (ἐκατ/εραξ τῶν) ἐντὶ καὶ ἀπεναντιον γωνῶν μείζων ἐστίν.

)/εστω τριγωνον ταβγ, καὶ προσεκβεβληθῶ αὐτοῦ μίᾳ πλευρῇ (ἡ βγ) ἐπὶ δ; λ, (/οτι (ἡ) ἐκτῶν/ια (ἡ) ὑπαγδ μείζων) ἐστὶ (ἐκατ/εραξ τῶν) ἐντὶ καὶ ἀπεναντιον τῶν (ὑπγβα, βαγ γωνῶν).

τεμνησθῶ (ἡ) αὐγ διῖθα καττε, καὶ ἐπιζευθθεῖσιν (ἡ) βε) ἐκβεβλησθῶ ἐπὶ εὐθείᾳ ἐπὶ ζ, καὶ κείσθω τῇ βε) ἰση (ἡ) εζ, καὶ ἐπεζεύθθω (ἡ) ζγ, καὶ διήθθω (ἡ) αὐγ) ἐπὶ τη.

)/επεὶ ὁ=υν) ἰση) ἐστὶ (ἡ) μαε τῇ εγ, (ἡ) δβε τῇ εζ, δ/υο δᾶι αε, εβ δυστάξι γε, εζ) ἰσαι εἰς (ἐκατ/ερα (ἐκατ/ερα; καὶ γων/ια (ἡ) ὑπαεβ γων/ια τῇ (ὑπζεγ) ἰση) ἐστίν; κατκορυφγ/αρ; β/ασιξ) /αρα (ἡ) αβ β/ασει τῇ ζγ) ἰση) ἐστίν, καὶ ταβε τριγωνον τ=ω ζεγ τριγωνῳ) ἐστὶ ἰσον, καὶ αἱ λοιπαὶ γων/ιαὶ τᾶξι λοιπᾶξι γων/ιαῖς) ἰσαι εἰς (ἐκατ/ερα (ἐκατ/ερα, (υφ) (αἱ) ἰσαι πλευραὶ (ὑποτένουσιν;)/ ἰση) /αρα) ἐστὶ (ἡ) ὑπβαε τῇ (ὑπεγζ. μείζων δ/ε) ἐστὶν (ἡ) ὑπεγδ τῇ ξ (ὑπεγζ; μείζων) /αρα (ἡ) ὑπαγδ τῇ ξ (ὑπβαε. (ομοίωξ δτῇ ξ βγ τεμνημ/ενῇ ξ διῖθα δειθθ/ησεται καὶ (ἡ) ὑπβγῇ, τουτ/ἐστὶν (ἡ) ὑπαγδ, μείζων καὶ τῇ ξ (ὑπαβγ).

παντ) /αρα τριγωνου μιᾶς τῶν πλευρῶν προσεκβληθείσης (ἡ) ἐκτῶν/ια (ἐκατ/εραξ τῶν) ἐντὶ καὶ ἀπεναντιον γωνῶν μείζων ἐστίν; (/οπερ) /εδει δεῖχαι.

Proposition 16

For any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is greater than each of the internal and opposite angles, CBA and BAC .

Let the (straight-line) AC be cut in half at (point) E [Prop. 1.10]. And BE being joined, let it be produced in a straight-line to (point) F .[†] And let EF be made equal to BE [Prop. 1.3], and let FC be joined, and let AC be drawn through to (point) G .

Therefore, since AE is equal to EC , and BE to EF , the two (straight-lines) AE , EB are equal to the two (straight-lines) CE , EF , respectively. Also, angle AEB is equal to angle FEC , for (they are) vertically opposite [Prop. 1.15]. Thus, the base AB is equal to the base FC , and the triangle

ABE is equal to the triangle FEC , and the remaining angles subtended by the equal sides are equal to the corresponding remaining angles [Prop. 1.4]. Thus, BAE is equal to ECF . But ECD is greater than ECF . Thus, ACD is greater than BAE . Similarly, by having cut BC in half, it can be shown (that) BCG —that is to say, ACD —(is) also greater than ABC .

Thus, for any triangle, when one of the sides is produced, the external angle is greater than each of the internal and opposite angles. (Which is) the very thing it was required to show. [†] The implicit assumption that the point F lies in the interior of the angle ABC should be counted as an additional postulate.

17.

πανττριγωνου αι δ/υο γων/ιαι δ/υο)ορθ=ων)ελ/ασσον/εξ ε/ισι π/αντ=η μεταλαμβαν/ομεναι.
)/εστω τριγωνον ταβγ; λεγω, (/οτι το=υ αβγ τριγωνου αι δ/υο γων/ιαι δ/υο)ορθ=ων
)ελ/αττον/εξ ε/ισι π/αντη μεταλαμβαν/ομεναι.
)εκβεβλησθω γ(η βγ)επιδ.
 και)επε/ι τριγωνου το=υ αβγ)εκτ/οξ)εστι γων/ια (η (υπαγδ, μ/ειζων)εσττ=ηξ)εντκαι
)απεναντ/ιον τ=ηξ (υπαβγ. κοινπροσκεισθω (η (υπαγβ; αι)/αρα (υπαγδ, αγβ τ=ων
 (υπαβγ, βγα μ/ειζων/εξ ε/ισιν.)αλλ)αι (υπαγδ, αγβ δ/υο)ορθαιξ)/ισαι ε/ις/ιν; αι)/αρα
 (υπαβγ, βγα δ/υο)ορθ=ων)ελ/ασσον/εξ ε/ισιν. (ομ/οιωξ δδ/ειχομεν, (/οτι και αι (υπαγ,
 αγβ δ/υο)ορθ=ων)ελ/ασσον/εξ ε/ισι και)/ετι αι (υπαγβ, αβγ.

παντ)/αρα τριγωνου αι δ/υο γων/ιαι δ/υο)ορθ=ων)ελ/ασσον/εξ ε/ισι π/αντ=η μεταλαμβαν/ομεναι
 (/οπερ)/εδει δε=ιχαι.

Proposition 17

For any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles.

Let ABC be a triangle. I say that (the sum of) two angles of triangle ABC taken together in any (possible way) is less than two right-angles.

For let BC be produced to D .

And since the angle ACD is external to triangle ABC , it is greater than the internal and opposite angle ABC [Prop. 1.16]. Let ACB be added to both. Thus, the (sum of the angles) ACD and ACB is greater than the (sum of the angles) ABC and BCA . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ABC and BCA is less than two right-angles. Similarly, we can show that (the sum of) BAC and ACB is also less than two right-angles, and further (that the sum of) CAB and ABC (is less than two right-angles).

Thus, for any triangle, (the sum of) two angles taken together in any (possible way) is less than two right-angles. (Which is) the very thing it was required to show.

18.

πανττριγωνου (η μ/ειζων πλευρμ/ειζονα γων/ιαν (υποτεινει.
)/εστω γτριγωνον ταβγ μ/ειζονα)/εθον ταγ πλευρτ=ηξ αβ; λεγω, (/οτι και γων/ια
 (η (υπαβγ μ/ειζων)εσττ=ηξ (υπβγα.

)επεί γμείζων)εστ(η αγ τ=ηξ αβ, κείσθω τ=η αβ)/ιση (η αδ, καὶ)επεξέυθθω (η βδ.
 καὶ)επεί τριγ/ωνου το=υ βγδ)εκτ/οξ)εστι γων/ια (η (υπαδβ, μείζων)εσττ=ηξ)εντκαὶ
)απεναντ/ιον τ=ηξ (υπδγβ;)/ιση δ(η (υπαδβ τ=η (υπαβδ,)επεί καὶ πλευρ(η αβ τ=η αδ
)εστιν)/ιση; μείζων)/αρα καὶ (η (υπαβδ τ=ηξ (υπαγβ; πολλ=ω)/αρα (η (υπαβγ μείζων
)εσττ=ηξ (υπαγβ.

παντ)/αρα τριγ/ωνου (η μείζων πλευρτμείζονα γων/ιαν (υποτέινει; (/οπερ)/εδει
 δε=ιχαί.

Proposition 18

In any triangle, the greater side subtends the greater angle.

For let ABC be a triangle having side AC greater than AB . I say that angle ABC is also greater than BCA .

For since AC is greater than AB , let AD be made equal to AB [Prop. 1.3], and let BD be joined.

And since angle ADB is external to triangle BCD , it is greater than the internal and opposite (angle) DCB [Prop. 1.16]. But ADB (is) equal to ABD , since side AB is also equal to side AD [Prop. 1.5]. Thus, ABD is also greater than ACB . Thus, ABC is much greater than ACB .

Thus, in any triangle, the greater side subtends the greater angle. (Which is) the very thing it was required to show.

19.

πανττριγ/ωνου (υπτμείζονα γων/ιαν (η μείζων πλευρ(υποτέινει.

)/εστω τριγ/ωνον ταβγ μείζονα)/εθον τ(υπαβγ γων/ιαν τ=ηξ (υπβγα; λεγω, (/οτι
 καὶ πλευρ(η αγ πλευρ=αξ τ=ηξ αβ μείζων)εστ/ιν.

έι γμ/η,)/ητοι)/ιση)εστ(η αγ τ=η αβ))ελ/ασσων;)/ιση μόν=υν όυκ)/εστιν (η αγ τ=η
 αβ;)/ιση γ))=ην καὶ γων/ια (η (υπαβγ τ=η (υπαγβ; όυκ)/εστι δ/ε; όυκ)/αρα)/ιση)εστ(η
 αγ τ=η αβ. όυδμ)ελ/ασσων)εστ(η αγ τ=ηξ αβ;)ελ/ασσων γ))=ην καὶ γων/ια (η (υπαβγ
 τ=ηξ (υπαγβ; όυκ)/εστι δ/ε; όυκ)/αρα)ελ/ασσων)εστ(η αγ τ=ηξ αβ.)εδείθθη δ/ε, (/οτι
 όυδ)/ιση)εστ/ιν. μείζων)/αρα)εστ(η αγ τ=ηξ αβ.

παντ)/αρα τριγ/ωνου (υπτμείζονα γων/ιαν (η μείζων πλευρ(υποτέινει; (/οπερ)/εδει
 δε=ιχαί.

Proposition 19

In any triangle, the greater angle is subtended by the greater side.

Let ABC be a triangle having the angle ABC greater than BCA . I say that side AC is also greater than side AB .

For if not, AC is certainly either equal to, or less than, AB . In fact, AC is not equal to AB . For then angle ABC would also be equal to ACB [Prop. 1.5]. But it is not. Thus, AC is not equal to AB . Neither, indeed, is AC less than AB . For then angle ABC would also be less than ACB [Prop. 1.18]. But it is not. Thus, AC is not less than AB . But it was shown that (AC) is not equal (to AB) either. Thus, AC is greater than AB .

Thus, in any triangle, the greater angle is subtended by the greater side. (Which is) the very thing it was required to show.

20.

πανττριγ/ωνου ἂι δ/υο πλευρὰι τ=ηξ λοιπ=ηξ μείζον/εξ ἔισι π/αντη μεταλαμβάν/ομεναι.

)/εστω γτρίγωνον ταβγ; λέγω, (/οτι το=υ αβγ τριγ/ωνου ἂι δ/υο πλευρὰι τ=ηξ λοιπ=ηξ μείζον/εξ ἔισι π/αντη μεταλαμβάν/ομεναι, ἂι μβα, αγ τ=ηξ βγ, ἂι δαβ, βγ τ=ηξ αγ, ἂι δβγ, γα τ=ηξ αβ.

δίηθω γ(η βα)επτδ σημειον, καὶ κείσθω τ=η γα)/ιση (η αδ, καὶ)επεζεύθω (η δγ.

)επὲι ὁ=υν)/ιση)εστ(η δα τ=η αγ,)/ιση)εστκαὶ γων/ια (η (υπαδγ τ=η (υπαγδ; μείζων)/αρα (η (υπβγδ τ=ηξ (υπαδγ; καὶ)επὲι τριγ/ων/ον)εστι τδγβ μείζονα)/εθον τ(υπβγδ γων/ιαν τ=ηξ (υπβδγ, (υπδτμείζονα γων/ιαν (η μείζων πλευρ(υποτείνει, (η δβ)/αρα τ=ηξ βγ)εστι μείζων.)/ιση δ(η δα τ=η αγ; μείζονεξ)/αρα ἂι βα, αγ τ=ηξ βγ; (ομοίωξ δδείχομεν, (/οτι καὶ ἂι μαβ, βγ τ=ηξ γα μείζον/εξ ἔισιν, ἂι δβγ, γα τ=ηξ αβ.

παντ)/αρα τριγ/ωνου ἂι δ/υο πλευρὰι τ=ηξ λοιπ=ηξ μείζον/εξ ἔισι π/αντη μεταλαμβάν/ομεναι; (/οπερ)/εδει δε=ιχαί.

Proposition 20

In any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side).

For let ABC be a triangle. I say that in triangle ABC (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (So), (the sum of) BA and AC (is greater) than BC , (the sum of) AB and BC than AC , and (the sum of) BC and CA than AB .

For let BA be drawn through to point D , and let AD be made equal to CA [Prop. 1.3], and let DC be joined.

Therefore, since DA is equal to AC , the angle ADC is also equal to ACD [Prop. 1.5]. Thus, BCD is greater than ADC . And since DCB is a triangle having the angle BCD greater than BDC , and the greater angle subtends the greater side [Prop. 1.19], DB is thus greater than BC . But DA is equal to AC . Thus, (the sum of) BA and AC is greater than BC . Similarly, we can show that (the sum of) AB and BC is also greater than CA , and (the sum of) BC and CA than AB .

Thus, in any triangle, (the sum of) two sides taken together in any (possible way) is greater than the remaining (side). (Which is) the very thing it was required to show.

21.

)ἐὰν τριγ/ωνου)επιμῖαξ τ=ων πλευρ=ων)απτ=ων περ/ατων δ/υο ἐυθε=ιαι)εντισταθ=ωσιν, αἱ συσταθε=ισαι τ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων)ελαττονεξ μ)/εσονται, μείζονα δγων/ιαν περιέχουσιν.

τριγ/ωνου γτο=υ αβγ)επιμῖαξ τ=ων πλευρ=ων τ=ηξ βγ)απτ=ων περ/ατων τ=ων β, γ δ/υο ἐυθε=ιαι)εντσυνεστ/ατωσαν αἱ βδ, δγ; λεγω, (/οτι αἱ βδ, δγ τ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων τ=ων βα, αγ)ελασσονεξ μ/εν εἰσιν, μείζονα δγων/ιαν περιέθουσι τ(υπβδγ τ=ηξ (υπβαγ.

δίηθθω γ(ηβδ)επτε. καὶ)επεί πανττριγ/ωνου αἱ δ/υο πλευρὰι τ=ηξ λοιπ=ηξ μείζον/εξ εἰσιν, το=υ αβε)/αρα τριγ/ωνου αἱ δ/υο πλευρὰι αἱ αβ, αε τ=ηξ βε μείζον/εξ εἰσιν; κοινπροσκέισθω (η εγ; αἱ)/αρα βα, αγ τ=ων βε, εγ μείζον/εξ εἰσιν. π/αλιν,)επεί το=υ γεδ τριγ/ωνου αἱ δ/υο πλευρὰι αἱ γε, εδ τ=ηξ γδ μείζον/εξ εἰσιν, κοινπροσκέισθω (η δβ; αἱ γε, εβ)/αρα τ=ων γδ, δβ μείζον/εξ εἰσιν.)αλλτ=ων βε, εγ μείζονεξ)εδείθθησαν αἱ βα, αγ; πολλ=ω)/αρα αἱ βα, αγ τ=ων βδ, δγ μείζον/εξ εἰσιν.

π/αλιν,)επεί πανττριγ/ωνου (η)εκτγων/ια τ=ηξ)εντκαὶ)απεναντ/ιον μείζων)εστ/ιν, το=υ γδε)/αρα τριγ/ωνου (η)εκτγων/ια (η (υπβδγ μείζων)εσττ=ηξ (υπγεδ. διὰ τα)υττόινυν καὶ το=υ αβε τριγ/ωνου (η)εκτγων/ια (η (υπγεβ μείζων)εσττ=ηξ (υπβαγ.)αλλτ=ηξ (υπγεβ μείζων)εδείθθη (η (υπβδγ; πολλ=ω)/αρα (η (υπβδγ μείζων)εσττ=ηξ (υπβαγ.

)ἐὰν)/αρα τριγ/ωνου)επιμῖαξ τ=ων πλευρ=ων)απτ=ων περ/ατων δ/υο ἐυθε=ιαι)εντισταθ=ωσιν, αἱ συσταθε=ισαι τ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων)ελαττονεξ μ/εν εἰσιν, μείζονα δγων/ιαν περιέθουσιν; (/οπερ)/εδει δε=ιχαί.

Proposition 21

If two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) will be less than the two remaining sides of the triangle, but will encompass a greater angle.

For let the two internal straight-lines BD and DC be constructed on one of the sides BC of the triangle ABC , from its ends B and C (respectively). I say that BD and DC are less than the (sum of the) two remaining sides of the triangle BA and AC , but encompass an angle BDC greater than BAC .

For let BD be drawn through to E . And since in any triangle (the sum of any) two sides is greater than the remaining (side) [Prop. 1.20], in triangle ABE the (sum of the) two sides AB and AE is thus greater than BE . Let EC be added to both. Thus, (the sum of) BA and AC is greater than (the sum of) BE and EC . Again, since in triangle CED the (sum of the) two sides CE and ED is greater than CD , let DB be added to both. Thus, (the sum of) CE and EB is greater than (the sum of) CD and DB . But, (the sum of) BA and AC was shown (to be) greater than (the sum of) BE and EC . Thus, (the sum of) BA and AC is much greater than (the sum of) BD and DC .

Again, since in any triangle the external angle is greater than the internal and opposite (angles) [Prop. 1.16], in triangle CDE the external angle BDC is thus greater than CED . Accordingly, for the same (reason), the external angle CEB of the triangle ABE is also greater than BAC . But, BDC was shown (to be) greater than CEB . Thus, BDC is much greater than BAC .

Thus, if two internal straight-lines are constructed on one of the sides of a triangle, from its ends, the constructed (straight-lines) are less than the two remaining sides of the triangle, but encompass a greater angle. (Which is) the very thing it was required to show.

(ἐκ τριῶν ἐυθεῖων, αἱ εἰσιν) /ισαὶ τριῖταιξ δοθείσαιξ [ἐυθείαιξ], τρ/ιγωνον συστ/ησασθαι; δε=ι δτδ/υο τ=ηξ λοιπ=ηξ μείζοναξ εἶναι π/αντη μεταλαμβανομ/εναξ [διὰ τὰ καὶ πανττριγ/ωνου τδ/υο πλευρτ=ηξ λοιπ=ηξ μείζοναξ εἶναι π/αντη μεταλαμβανομ/εναξ].

/εστωσαν αἱ δοθε=ισαὶ τρε=ιξ ἐυθε=ιαὶ αἱ α, β, γ, (=ων αἱ δ/υο τ=ηξ λοιπ=ηξ μείζονεξ) /εστωσαν π/αντη μεταλαμβαν/ομεναι, αἱ μα, β τ=ηξ γ, αἱ δα, γ τ=ηξ β, καὶ /ετι αἱ β, γ τ=ηξ α; δε=ι δ) ἐκ τ=ων /ισων τᾶιξ α, β, γ τρ/ιγωνον συστ/ησασθαι.

(ἐκκέισθω τιξ ἐυθε=ια (ἡ δε πεπερασμ/ενη μακτδ) /απειροξ δκαττε, καὶ κείσθω τ=η μα) /ιση (ἡ δζ, τ=η δβ) /ιση (ἡ ζη, τ=η δγ) /ιση (ἡ ηθ; καὶ κ/εντρῳ μτ=ω ζ, διαστ/ηματι δτ=ω ζδ κ/υκλοξ γεγρ/αφθω (ο δκλ; π/άλιν κ/εντρῳ μτ=ω η, διαστ/ηματι δτ=ω ηθ κ/υκλοξ γεγρ/αφθω (ο κλθ, καὶ) ἐπεξέυθθωσαν αἱ κζ, κη; λ/εγω, (/οτι) ἐκ τριῶν ἐυθεῖων τ=ων /ισων τᾶιξ α, β, γ τρ/ιγωνον συν/εσταται τκζη.

/επὲι γτζ σημε=ιον κ/εντρον) εστο=υ δκλ κ/υκλου, (/ιση) εστ(ἡ ζδ τ=η ζκ;) αλλ(ἡ ζδ τ=η α) εστιν) /ιση. καὶ (ἡ κζ) /αρα τ=η α) εστιν) /ιση. π/άλιν, /επὲι τῇ σημε=ιον κ/εντρον) εστο=υ λκθ κ/υκλου, (/ιση) εστ(ἡ ηθ τ=η ηκ;) αλλ(ἡ ηθ τ=η γ) εστιν) /ιση; καὶ (ἡ κη) /αρα τ=η γ) εστιν) /ιση.)εσδκαὶ (ἡ ζη τ=η β) /ιση; αἱ τρε=ιξ) /αρα ἐυθε=ιαὶ αἱ κζ, ζη, ηκ τριῖταιξ α, β, γ) /ισαὶ εἰς/ιν.

(ἐκ τριῶν) /αρα ἐυθεῖων τ=ων κζ, ζη, ηκ, αἱ εἰσιν) /ισαὶ τριῖταιξ δοθείσαιξ ἐυθείαιξ τᾶιξ α, β, γ, τρ/ιγωνον συν/εσταται τκζη; (/οπερ) /εδει ποιῆσαι.

Proposition 22

To construct a triangle from three straight-lines which are equal to three given [straight-lines]. It is necessary for (the sum of) two (of the straight-lines) taken together in any (possible way) to be greater than the remaining (one), [on account of the (fact that) in any triangle (the sum of) two sides taken together in any (possible way) is greater than the remaining (one) [Prop. 1.20]].

Let A , B , and C be the three given straight-lines, of which let (the sum of) two taken together in any (possible way) be greater than the remaining (one). (Thus), (the sum of) A and B (is greater) than C , (the sum of) A and C than B , and also (the sum of) B and C than A . So it is required to construct a triangle from (straight-lines) equal to A , B , and C .

Let some straight-line DE be set out, terminated at D , and infinite in the direction of E . And let DF made equal to A , and FG equal to B , and GH equal to C [Prop. 1.3]. And let the circle DKL be drawn with center F and radius FD . Again, let the circle KLH be drawn with center G and radius GH . And let KF and KG be joined. I say that the triangle KFG has been constructed from three straight-lines equal to A , B , and C .

For since point F is the center of the circle DKL , FD is equal to FK . But, FD is equal to A . Thus, KF is also equal to A . Again, since point G is the center of the circle LKH , GH is equal to GK . But, GH is equal to C . Thus, KG is also equal to C . And FG is also equal to B . Thus, the three straight-lines KF , FG , and GK are equal to A , B , and C (respectively).

Thus, the triangle KFG has been constructed from the three straight-lines KF , FG , and GK , which are equal to the three given straight-lines A , B , and C (respectively). (Which is) the very thing it was required to do.

23.

πρτ=η δοθείση ἐυθεία καὶ τ=ω πρ(α)υτ=η σημείω τ=η δοθείση γων/ια ἐυθυγρ/αμμω)/ισην γων/ιαν ἐυθ/υγραμμον συστ/ησασθαι.

)εστω (η μδοθε=ισα ἐυθε=ια (η αβ, τδπρ(α)υτ=η σημει=ιον τα, (η δδοθε=ισα γων/ια ἐυθ/υγραμμοξ (η (υπδγε; δε=ι δπρτ=η δοθείση ἐυθεία τ=η αβ καὶ τ=ω πρ(α)υτ=η σημείω τ=ω α τ=η δοθείση γων/ια ἐυθυγρ/αμμω τ=η (υπδγε)/ισην γων/ιαν ἐυθ/υγραμμον συστ/ησασθαι.

ἐιλ/ηφθω)εφ) (εκατ/εραξ τ=ων γδ, γε τυθ/οντα σημει=ια τδ, ε, καὶ)επεξέυθθω (η δε; καὶ)εκ τριῶν ἐυθειῶν, ἅι εἰσιν)/ισαι τριςτᾶιξ γδ, δε, γε, τρ/ιγωνον συνεστ/ατω ταζη, (/ωστε)/ισην εἶναι τμγδ τ=η αζ, τδγε τ=η αη, καὶ)/ετι τδε τ=η ζη.

)επὲι ὁ=υν δ/υο αἱ δγ, γε δ/υο τᾶιξ ζα, αη)/ισαι εἰς(εκατ/ερα (εκατ/ερα, καὶ β/ασιξ (η δε β/ασει τ=η ζη)/ιση, γων/ια)/αρα (η (υπδγε γων/ια τ=η (υπζαη)εστιν)/ιση.

πρ)/αρα τ=η δοθείση ἐυθεία τ=η αβ καὶ τ=ω πρ(α)υτ=η σημείω τ=ω α τ=η δοθείση γων/ια ἐυθυγρ/αμμω τ=η (υπδγε)/ιση γων/ια ἐυθ/υγραμμοξ συν/εσταται (η (υπζαη; (/οπερ)/εδει ποιῆσαι.

Proposition 23

To construct a rectilinear angle equal to a given rectilinear angle at a (given) point on a given straight-line.

Let AB be the given straight-line, A the (given) point on it, and DCE the given rectilinear angle. So it is required to construct a rectilinear angle equal to the given rectilinear angle DCE at the (given) point A on the given straight-line AB .

Let the points D and E be taken at random on each of the (straight-lines) CD and CE (respectively), and let DE be joined. And let the triangle AFG be constructed from three straight-lines which are equal to CD , DE , and CE , such that CD is equal to AF , CE to AG , and further DE to FG [Prop. 1.22].

Therefore, since the two (straight-lines) DC , CE are equal to the two (straight-lines) FA , AG , respectively, and the base DE is equal to the base FG , the angle DCE is thus equal to the angle FAG [Prop. 1.8].

Thus, the rectilinear angle FAG , equal to the given rectilinear angle DCE , has been constructed at the (given) point A on the given straight-line AB . (Which is) the very thing it was required to do.

24.

)ἐαν δ/υο τρ/ιγωνα τδ/υο πλευρ[τᾶιξ] δ/υο πλευρᾶιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα, τδγων/ιαν τ=ηξ γων/ιαξ μείζονα)/εθῃ τ(υπτ=ων)/ισων ἐυθειῶν περιεθომ/ενην, καὶ τβ/ασιν τ=ηξ β/ασεωξ μείζονα)/εχει.

)εστω δ/υο τρ/ιγωνα ταβγ, δεζ τδ/υο πλευρταβ, αγ τᾶιξ δ/υο πλευρᾶιξ τᾶιξ δε, δζ)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα, τμαβ τ=η δε τδαγ τ=η δζ, (η δπρτ=ω α γων/ια τ=ηξ πρτ=ω δ γων/ιαξ μείζων)/εστω; λ/εγω, (/οτι καὶ β/ασιξ (η βγ β/ασεωξ τ=ηξ εζ μείζων)εστ/ιν.

)επὲι γμείζων (η (υπβαγ γων/ια τ=ηξ (υπεδζ γων/ιαξ, συνεστ/ατω πρτ=η δε ἐυθεία καὶ τ=ω πρ(α)υτ=η σημείω τ=ω δ τ=η (υπβαγ γων/ια)/ιση (η (υπεδη, καὶ κείσθω (οποτ/ερα τ=ων αγ, δζ)/ιση (η δη, καὶ)επεξέυθθωσαν αἱ εη, ζη.

)/επεί ὁ=υν)/ιση)εστ(η μαβ τ=η δε, (η δαγ τ=η δη, δ/υο δάι βα, αγ δυσταῖξ εδ, δη)/ισαι
 εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ια (η (υπβαγ γων/ια τ=η (υπεδη)/ιση; β/ασιξ)/αρα (η
 βγ β/ασει τ=η εη)εστιν)/ιση. π/αλιν,)επεί)/ιση)εστ(η δζ τ=η δη,)/ιση)εστκαὶ (η (υπδηζ
 γων/ια τ=η (υπδζη; μείζων)/αρα (η (υπδζη τ=ηξ (υπεηζ; πολλ=ω)/αρα μείζων)εστ(η
 (υπεζη τ=ηξ (υπεηζ. καὶ)επεί τρ/ιγων/ον)εστι τεζη μείζονα)/εθον τ(υπεζη γων/ιαν τ=ηξ
 (υπεηζ, (υπδτμείζονα γων/ιαν (η μείζων πλευρ(υποτέινει, μείζων)/αρα καὶ πλευρ(η εη
 τ=ηξ εζ.)/ιση δ(η εη τ=η βγ; μείζων)/αρα καὶ (η βγ τ=ηξ εζ.

)/εάν)/αρα δ/υο τρ/ιγωνα τδ/υο πλευρδυσπλευρᾶιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα,
 τδγων/ιαν τ=ηξ γων/ιαξ μείζονα)/εθῃ τ(υπτ=ων)/ισων ἐυθεῖων περιεθομ/ενην, καὶ
 τβ/ασιν τ=ηξ β/ασεωξ μείζονα (/εχει; (/οπερ)/εδει δε=ιχαί.

Proposition 24

If two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter).

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively. (That is), AB (equal) to DE , and AC to DF . Let them also have the angle at A greater than the angle at D . I say that the base BC is also greater than the base EF .

For since angle BAC is greater than angle EDF , let (angle) EDG , equal to angle BAC , be constructed at the point D on the straight-line DE [Prop. 1.23]. And let DG be made equal to either of AC or DF [Prop. 1.3], and let EG and FG be joined.

Therefore, since AB is equal to DE and AC to DG , the two (straight-lines) BA , AC are equal to the two (straight-lines) ED , DG , respectively. Also the angle BAC is equal to the angle EDG . Thus, the base BC is equal to the base EG [Prop. 1.4]. Again, since DF is equal to DG , angle DGF is also equal to angle DFG [Prop. 1.5]. Thus, DFG (is) greater than EGF . Thus, EFG is much greater than EGF . And since triangle EFG has angle EFG greater than EGF , and the greater angle is subtended by the greater side [Prop. 1.19], side EG (is) thus also greater than EF . But EG (is) equal to BC . Thus, BC (is) also greater than EF .

Thus, if two triangles have two sides equal to two sides, respectively, but (one) has the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the other), (then the former triangle) will also have a base greater than the base (of the latter). (Which is) the very thing it was required to show.

25.

)/εάν δ/υο τρ/ιγωνα τδ/υο πλευρδυσπλευρᾶιξ)/ισαξ)/εθῃ (εκατ/εραν (εκατ/ερα, τδβ/ασιν
 τ=ηξ β/ασεωξ μείζονα)/εθῃ, καὶ τγων/ιαν τ=ηξ γων/ιαξ μείζονα (/εχει τ(υπτ=ων)/ισων
 ἐυθεῖων περιεθομ/ενην.

)/εστω δ/υο τρ/ιγωνα ταβγ, δεζ τδ/υο πλευρταβ, αγ ταῖξ δ/υο πλευρᾶιξ ταῖξ δε, δζ
)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα, τμαβ τ=η δε, τδαγ τ=η δζ; β/ασιξ δ(η βγ β/ασεωξ
 τ=ηξ εζ μείζων)/εστω; λ/εγω, (/οτι καὶ γων/ια (η (υπβαγ γων/ιαξ τ=ηξ (υπεδζ μείζων
)εστ/ιν.

εἰ γμ/η,)/ητοι)/ιση)εστα)υτ=η))ελ/ασσων;)/ιση μὲ=υν οὐκ)/εστιν (η (υπβαγ τ=η
 (υπεδζ;)/ιση γ))=ην καὶ β/ασιξ (η βγ β/ασει τ=η εζ; οὐκ)/εστι δ/ε. οὐκ)/αρα)/ιση

)εστγων/ια (η (υπβαγ τ=η (υπεδζ; ουδμ)ελ/ασσων)εστ(η (υπβαγ τ=ηξ (υπεδζ;)ελ/ασσων γ))=ην και β/ασιξ (η βγ β/ασεωξ τ=ηξ εζ; ουκ)/εστι δ/ε; ουκ)/αρα)ελ/ασσων)εστ(η (υπβαγ γων/ια τ=ηξ (υπεδζ.)εδείθθη δ/ε, (/οτι ουδ)/ιση; μείζων)/αρα)εστ(η (υπβαγ τ=ηξ (υπεδζ.

)εαν)/αρα δ/υο τρ/ιγωνα τδ/υο πλευρδυσπλευρᾱιξ)/ισαξ)/εθη (εκατ/εραν (εκ/ατερᾱ, τδβας/ιν τ=ηξ β/ασεωξ μείζονα)/εθη, και τγων/ιαν τ=ηξ γων/ιαξ μείζονα (/εχει τ(υπτ=ων)/ισων ευθείων περιεθομ/ενην; (/οπερ)/εδει δε=ιχαι.

Proposition 25

If two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter).

Let ABC and DEF be two triangles having the two sides AB and AC equal to the two sides DE and DF , respectively (That is), AB (equal) to DE , and AC to DF . And let the base BC be greater than the base EF . I say that angle BAC is also greater than EDF .

For if not, (BAC) is certainly either equal to, or less than, (EDF). In fact, BAC is not equal to EDF . For then the base BC would also be equal to the base EF [Prop. 1.4]. But it is not. Thus, angle BAC is not equal to EDF . Neither, indeed, is BAC less than EDF . For then the base BC would also be less than the base EF [Prop. 1.24]. But it is not. Thus, angle BAC is not less than EDF . But it was shown that (BAC is) not equal (to EDF) either. Thus, BAC is greater than EDF .

Thus, if two triangles have two sides equal to two sides, respectively, but (one) has a base greater than the base (of the other), (then the former triangle) will also have the angle encompassed by the equal straight-lines greater than the (corresponding) angle (in the latter). (Which is) the very thing it was required to show.

26.

)εαν δ/υο τρ/ιγωνα τδ/υο γων/ιαξ δυσγων/ιαιξ)/ισαξ)/εθη (εκατ/εραν (εκατ/ερα και μ/ιαν πλευρμῖᾱ πλευρ=ᾱ)/ισην)/ητοι τπρτᾱιξ)/ισαιξ γων/ιαιξ)τ(υποτῑνουσαν (υπμ/ιαν τ=ων)/ισων γωνῖων, και τλοιππλευρτᾱιξ λοιπᾱιξ πλευρᾱιξ)/ισαξ (/εχει [(εκατ/εραν (εκατ/ερα] και τλοιπγων/ιαν τ=η λοιπ=η γων/ια.

)εστω δ/υο τρ/ιγωνα ταβγ, δεζ τδ/υο γων/ιαξ τ(υπαβγ, βγα δυστᾱιξ (υπδεζ, εζδ)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα, τμ(υπαβγ τ=η (υπδεζ, τδ(υπβγα τ=η (υπεζδ;)εθ/ετω δκαι μ/ιαν πλευρμῖᾱ πλευρ=ᾱ)/ισην, πρ/οτερον τπρτᾱιξ)/ισαιξ γων/ιαιξ τβγ τ=η εζ; λεγω, (/οτι και τλοιππλευρτᾱιξ λοιπᾱιξ πλευρᾱιξ)/ισαξ (/εχει (εκατ/εραν (εκατ/ερα, τμαβ τ=η δε τδαγ τ=η δζ, και τλοιπγων/ιαν τ=η λοιπ=η γων/ια, τ(υπβαγ τ=η (υπεδζ.

ῑ γ)/ανισ/οξ)εστιν (η αβ τ=η δε, μ/ια α)υτ=ων μείζων)εστ/ιν.)εστω μείζων (η αβ, και κείσθω τ=η δε)/ιση (η βη, και)επεξῑυθθω (η ηγ.

)επει ουν)/ιση)εστ(η μβη τ=η δε, (η δβγ τ=η εζ, δ/υο δᾱι βη, βγ δυστᾱιξ δε, εζ)/ισαι εῖς(εκατ/ερα (εκατ/ερα; και γων/ια (η (υπηβγ γων/ια τ=η (υπδεζ)/ιση)εστ/ιν; β/ασιξ)/αρα (η ηγ β/ασει τ=η δζ)/ιση)εστ/ιν, και τηβγ τρ/ιγωνον τ=ω δεζ τριγ/ωνω)/ισον

)/εστ/ιν, καὶ αἱ λοιπὰὶ γων/ιαὶ τᾶιξ λοιπᾶιξ γων/ιαὶξ)/ισαὶ)/εσονται, (υφ) (αἱ)/ισαὶ πλευρὰὶ (υποτέινουσιν;)/ιση)/αρα (η (υπηγβ γων/ια τ=η (υπδζε.)αλλ(η (υπδζε τ=η (υπβγα (υπ/οκειται)/ιση; καὶ (η (υπβγη)/αρα τ=η (υπβγα)/ιση)εστ/ιν, (η)ελλ/ασσων τ=η μείζονι; (/οπερ)αδ/υνατον. οὐκ)/αρα)/ανισ/οξ)εστιν (η αβ τ=η δε.)/ιση)/αρα.)/εστι δκαὶ (η βγ τ=η εζ)/ιση; δ/υο δαὶ αβ, βγ δυστᾶιξ δε, εζ)/ισαὶ εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ια (η (υπαβγ γων/ια τ=η (υπδεξ)εστιν)/ιση; β/ασιξ)/αρα (η αγ β/ασει τ=η δζ)/ιση)εστ/ιν, καὶ λοιπγων/ια (η (υπβαγ τ=η λοιπ=η γων/ια τ=η (υπεδξ)/ιση)εστ/ιν.

)αλλδπ/αλιν)/εστωσαν αἱ (υπτ)/ισαξ γων/ιαξ πλευρὰὶ (υποτέινουσαι)/ισαὶ, (ωξ (η αβ τ=η δε; λ/εγω π/αλιν, (/οτι καὶ αἱ λοιπὰὶ πλευρὰὶ τᾶιξ λοιπᾶιξ πλευρᾶιξ)/ισαὶ)/εσονται, (η μαγ τ=η δζ, (η δβγ τ=η εζ καὶ)/ετι (η λοιπγων/ια (η (υπβαγ τ=η λοιπ=η γων/ια τ=η (υπεδξ)/ιση)εστ/ιν.

εἰ γ)/ανισ/οξ)εστιν (η βγ τ=η εζ, μ/ια α)υτ=ων μείζων)εστ/ιν.)/εστω μείζων, εἰ δυνατ/ον, (η βγ, καὶ κείσθω τ=η εζ)/ιση (η βθ, καὶ)επεξέυθθω (η αθ. καὶ)επ)/ιση)εστ(η μβθ τ=η εζ (η δαβ τ=η δε, δ/υο δαὶ αβ, βθ δυστᾶιξ δε, εζ)/ισαὶ εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ιαξ)/ισαξ περίεθουσιν; β/ασιξ)/αρα (η αθ β/ασει τ=η δζ)/ιση)εστ/ιν, καὶ ταβθ τρ/ιγωνον τ=ω δεξ τριγ/ωνω)/ισον)εστ/ιν, καὶ αἱ λοιπὰὶ γων/ιαὶ τᾶιξ λοιπᾶιξ γων/ιαὶξ)/ισαὶ)/εσονται, (υφ) (αἱ)/ισαξ πλευρὰὶ (υποτέινουσιν;)/ιση)/αρα)εστ(η (υπβθα γων/ια τ=η (υπεξδ.)αλλ(η (υπεξδ τ=η (υπβγα)εστιν)/ιση; τριγ/ωνου δτο=υ αθγ (η)εκτγων/ια (η (υπβθα)/ιση)εσττ=η)εντκαὶ)απεναντ/ιον τ=η (υπβγα; (/οπερ)αδ/υνατον. οὐκ)/αρα)/ανισ/οξ)εστιν (η βγ τ=η εζ;)/ιση)/αρα.)εστδκαὶ (η αβ τ=η δε)/ιση. δ/υο δαὶ αβ, βγ δ/υο τᾶιξ δε, εζ)/ισαὶ εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ιαξ)/ισαξ περίεθουσι; β/ασιξ)/αρα (η αγ β/ασει τ=η δζ)/ιση)εστ/ιν, καὶ ταβγ τρ/ιγωνον τ=ω δεξ τριγ/ωνω)/ισον καὶ λοιπγων/ια (η (υπβαγ τ=η λοιπγων/ια τ=η (υπεδξ)/ιση.

)εαν)/αρα δ/υο τρ/ιγωνα τδ/υο γων/ιαξ δυσγων/ιαὶξ)/ισαξ)/εθη (εκατ/εραν (εκατ/ερα καὶ μ/ιαν πλευρμῖα πλευρ=α)/ισην)/ητοι τπρτᾶιξ)/ισαὶξ γων/ιαὶξ,)τ(υποτέινουσιν (υπμ/ιαν τ=ων)/ισων γωνῶν, καὶ τλοιππλευρτᾶιξ λοιπᾶιξ πλευρᾶιξ)/ισαξ)/εχει καὶ τλοιπγων/ιαν τ=η λοιπ=η γων/ια; (/οπερ)/εδει δε=ιχαί.

Proposition 26

If two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the [corresponding] remaining sides, and the remaining angle (equal) to the remaining angle.

Let ABC and DEF be two triangles having the two angles ABC and BCA equal to the two (angles) DEF and EFD , respectively. (That is) ABC (equal) to DEF , and BCA to EFD . And let them also have one side equal to one side. First of all, the (side) by the equal angles. (That is) BC (equal) to EF . I say that they will have the remaining sides equal to the corresponding remaining sides. (That is) AB (equal) to DE , and AC to DF . And (they will have) the remaining angle (equal) to the remaining angle. (That is) BAC (equal) to EDF .

For if AB is unequal to DE (then) one of them is greater. Let AB be greater, and let BG be made equal to DE [Prop. 1.3], and let GC be joined.

Therefore, since BG is equal to DE , and BC to EF , the two (straight-lines) GB , $BC^\dagger$ are equal to the two (straight-lines) DE , EF , respectively. And angle GBC is equal to angle DEF . Thus, the

base GC is equal to the base DF , and triangle GBC is equal to triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, GCB (is equal) to DFE . But, DFE was assumed (to be) equal to BCA . Thus, BCG is also equal to BCA , the lesser to the greater. The very thing (is) impossible. Thus, AB is not unequal to DE . Thus, (it is) equal. And BC is also equal to EF . So the two (straight-lines) AB, BC are equal to the two (straight-lines) DE, EF , respectively. And angle ABC is equal to angle DEF . Thus, the base AC is equal to the base DF , and the remaining angle BAC is equal to the remaining angle EDF [Prop. 1.4].

But, again, let the sides subtending the equal angles be equal: for instance, (let) AB (be equal) to DE . Again, I say that the remaining sides will be equal to the remaining sides. (That is) AC (equal) to DF , and BC to EF . Furthermore, the remaining angle BAC is equal to the remaining angle EDF .

For if BC is unequal to EF (then) one of them is greater. If possible, let BC be greater. And let BH be made equal to EF [Prop. 1.3], and let AH be joined. And since BH is equal to EF , and AB to DE , the two (straight-lines) AB, BH are equal to the two (straight-lines) DE, EF , respectively. And the angles they encompass (are also equal). Thus, the base AH is equal to the base DF , and the triangle ABH is equal to the triangle DEF , and the remaining angles subtended by the equal sides will be equal to the (corresponding) remaining angles [Prop. 1.4]. Thus, angle BHA is equal to EFD . But, EFD is equal to BCA . So, in triangle AHC , the external angle BHA is equal to the internal and opposite angle BCA . The very thing (is) impossible [Prop. 1.16]. Thus, BC is not unequal to EF . Thus, (it is) equal. And AB is also equal to DE . So the two (straight-lines) AB, BC are equal to the two (straight-lines) DE, EF , respectively. And they encompass equal angles. Thus, the base AC is equal to the base DF , and triangle ABC (is) equal to triangle DEF , and the remaining angle BAC (is) equal to the remaining angle EDF [Prop. 1.4].

Thus, if two triangles have two angles equal to two angles, respectively, and one side equal to one side—in fact, either that by the equal angles, or that subtending one of the equal angles—(then the triangles) will also have the remaining sides equal to the (corresponding) remaining sides, and the remaining angle (equal) to the remaining angle. (Which is) the very thing it was required to show. [†] The Greek text has “ BG, BC ”, which is obviously a mistake.

27.

)εἰς δ/υο ἐυθείας ἐυθεῖα)εμπ/ιπτουσα τ)εναλλγων/ιαξ)/ισαξ)αλλ/ηλαιξ ποῖη, παρ/αλληλοι)/εσονται)αλλ/ηλαιξ αἱ ἐυθεῖαι.

εἰς γδ/υο ἐυθείας ταβ, γδ ἐυθεῖα)εμπ/ιπτουσα (η εζ τ)εναλλγων/ιαξ τ(υπαεζ, εζδ)/ισαξ)αλλ/ηλαιξ ποιείτω; λεγω, (/οτι παρ/αλληλ/οξ)εστιν (η αβ τ=η γδ.

εἰ γμ/η,)εκβαλλ/ομεναι αἱ αβ, γδ συμπεσο=νται)/ητοι)επτβ, δ μ/ερη))επτα, γ.)εκβεβλ/ησθωσαν καὶ συμπιπτ/ετωσαν)επτβ, δ μ/ερη καττη. τριγ/ωνου δτο=υ ηεζ (η)εκγων/ια (η (υπαεζ)/ιση)εσττ=η)εντκαὶ)απεναντ/ιον τ=η (υπεζη; (/οπερ)εστ)αδ/υνατον; οὐκ)/αρα αἱ αβ, δγ)εκβαλλ/ομεναι συμπεσο=νται)επτβ, δ μ/ερη. (ομοίωξ δδειθθ/ησεται, (/οτι οὐδ)επτα, γ; αἱ δ)επμηδ/ετερα τμ/ερη συμπι/ιπτουσαι παρ/αλληλοί εἰσιν; παρ/αλληλοξ)/αρα)εστ(η αβ τ=η γδ.

)εἰς)/αρα εἰς δ/υο ἐυθείας ἐυθεῖα)εμπ/ιπτουσα τ)εναλλγων/ιαξ)/ισαξ)αλλ/ηλαιξ ποῖη, παρ/αλληλοι)/εσονται αἱ ἐυθεῖαι; (/οπερ)/εδει δε=ιχαι.

Proposition 27

If a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel to one another.

For let the straight-line EF , falling across the two straight-lines AB and CD , make the alternate angles AEF and EFD equal to one another. I say that AB and CD are parallel.

For if not, being produced, AB and CD will certainly meet together: either in the direction of B and D , or (in the direction) of A and C [Def. 1.23]. Let them be produced, and let them meet together in the direction of B and D at (point) G . So, for the triangle GEF , the external angle AEF is equal to the interior and opposite (angle) EFG . The very thing is impossible [Prop. 1.16]. Thus, being produced, AB and CD will not meet together in the direction of B and D . Similarly, it can be shown that neither (will they meet together) in (the direction of) A and C . But (straight-lines) meeting in neither direction are parallel [Def. 1.23]. Thus, AB and CD are parallel.

Thus, if a straight-line falling across two straight-lines makes the alternate angles equal to one another (then) the (two) straight-lines will be parallel (to one another). (Which is) the very thing it was required to show.

28.

ἔαν εἷξ δ/υο ἐυθείαξ ἐυθε=ια)εμπ/ιπτουσα τ)εκτγων/ιαν τ=η)εντκαὶ)απεναντ/ιον καὶ)επτα)υτμ/ερη)/ισην ποῖη)τ)εντκαὶ)επτα)υτμ/ερη δυσ)ορθᾶιξ)/ισαξ, παρ/αλληλοι)/εσονται)αλλ/ηλαιξ αἱ ἐυθε=ιαι.

εἷξ γδ/υο ἐυθείαξ ταβ, γδ ἐυθε=ια)εμπ/ιπτουσα (η εζ τ)εκτγων/ιαν τ(υπεηβ τ=η)εντκαὶ)απεναντ/ιον γων/ια τ=η (υπηθδ)/ισην ποιείτω)τ)εντκαὶ)επτα)υτμ/ερη τ(υπεηθ, ηθδ δυσ)ορθᾶιξ)/ισαξ; λεγω, (/οτι παρ/αλληλ/οξ)εστιν (η αβ τ=η γδ.

)επεὶ γ)/ιση)εστ(η (υπεηβ τ=η (υπηθδ,)αλλ(η (υπεηβ τ=η (υπαηθ)εστιν)/ιση, καὶ (η (υπαηθ)/αρα τ=η (υπηθδ)εστιν)/ιση; καὶ εἰσιν)εναλλ/αχ; παρ/αλληλοξ)/αρα)εστ(η αβ τ=η γδ.

π/αλιν,)επεὶ αἱ (υπεηθ, ηθδ δ/υο)ορθᾶιξ)/ισαὶ εἰς/ιν, εἰςδκαὶ αἱ (υπαηθ, βηθ δυσ)ορθᾶιξ)/ισαὶ, αἱ)/αρα (υπαηθ, βηθ τᾶιξ (υπεηθ, ηθδ)/ισαὶ εἰς/ιν; κοιν)αφηρ/ησθω (η (υπεηθ; λοιπ)/αρα (η (υπαηθ λοιπ=η τ=η (υπηθδ)εστιν)/ιση; καὶ εἰσιν)εναλλ/αχ; παρ/αλληλοξ)/αρα)εστ(η αβ τ=η γδ.

ἔαν)/αρα εἷξ δ/υο ἐυθείαξ ἐυθε=ια)εμπ/ιπτουσα τ)εκτγων/ιαν τ=η)εντκαὶ)απεναντ/ιον καὶ)επτα)υτμ/ερη)/ισην ποῖη)τ)εντκαὶ)επτα)υτμ/ερη δυσ)ορθᾶιξ)/ισαξ, παρ/αλληλοι)/εσονται αἱ ἐυθε=ιαὶ; (/οπερ)εδει δε=ιχαὶ.

Proposition 28

If a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the same side equal to two right-angles, (then) the (two) straight-lines will be parallel to one another.

For let EF , falling across the two straight-lines AB and CD , make the external angle EGB equal to the internal and opposite angle GHD , or the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles. I say that AB is parallel to CD .

For since (in the first case) EGB is equal to GHD , but EGB is equal to AGH [Prop. 1.15], AGH is thus also equal to GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

Again, since (in the second case, the sum of) BGH and GHD is equal to two right-angles, and (the sum of) AGH and BGH is also equal to two right-angles [Prop. 1.13], (the sum of) AGH and BGH is thus equal to (the sum of) BGH and GHD . Let BGH be subtracted from both. Thus, the remainder AGH is equal to the remainder GHD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

Thus, if a straight-line falling across two straight-lines makes the external angle equal to the internal and opposite angle on the same side, or (makes) the (sum of the) internal (angles) on the same side equal to two right-angles, (then) the (two) straight-lines will be parallel (to one another). (Which is) the very thing it was required to show.

29.

(η εἰς τπαραλλ/ηλουξ ἐυθείαξ ἐυθε=ια)εμπ/ιπτουσα τ/αξ τε)εναλλγων/ιαξ)/ισαξ)αλλ/ηλαιξ ποιε=ι καὶ τ)εκττ=η)εντκαὶ)απεναντ/ιον)/ισην καὶ τ)εντκαὶ)επτα)υτμ/ερη δυσ)ορθᾶιξ)/ισαξ.

εἰς γπαραλλ/ηλουξ ἐυθείαξ ταβ, γδ ἐυθε=ια)εμπιπτ/ετω (η εζ; λ/εγω, (/οτι τ)εναλλγων/ιαξ τ(υπαηθ, ηθδ)/ισαξ ποιε=ι καὶ τ)εκτγ/ων/ιαν τ(υπεηβ τ=η)εντκαὶ)απεναντ/ιον τ=η (υπηθδ)/ισην καὶ τ)εντκαὶ)επτα)υτμ/ερη τ(υπβηθ, ηθδ δυσ)ορθᾶιξ)/ισαξ.

εἰ γ)/ανις/οξ)εστιν (η (υπαηθ τ=η (υπηθδ, μ/ια α)υτ=ων μείζων)εστ/ιν.)/εστω μείζων (η (υπαηθ; κοινπροσκέισθω (η (υπβηθ; αἰ)/αρα (υπαηθ, βηθ τ=ων (υπβηθ, ηθδ μείζων/εξ εἰσιν.)αλλᾶι (υπαηθ, βηθ δυσ)ορθᾶιξ)/ισαι εἰς/ιν. [καὶ] αἰ)/αρα (υπβηθ, ηθδ δ/υο)ορθ=ων)ελ/ασσον/εξ εἰσιν. αἰ δ)απ))ελας/ονων)δ/υο)ορθ=ων)εκβαλλ/ομεναι εἰξ)/απειρον συμπ/ιπτουσιν; αἰ)/αρα αβ, γδ)εκβαλλ/ομεναι εἰξ)/απειρον συμπεσο=υνται; οὐ συμπ/ιπτουσι δδῖα τπαραλλ/ηλουξ α)υτ(υποκε=ισθαι; οὐκ)/αρα)/ανις/οξ)εστιν (η (υπαηθ τ=η (υπηθδ;)/ιση)/αρα.)αλλ(η (υπαηθ τ=η (υπεηβ)εστιν)/ιση; καὶ (η (υπεηβ)/αρα τ=η (υπηθδ)εστιν)/ιση; κοινπροσκέισθω (η (υπβηθ; αἰ)/αρα (υπεηβ, βηθ τᾶιξ (υπβηθ, ηθδ)/ισαι εἰς/ιν.)αλλᾶι (υπεηβ, βηθ δ/υο)ορθᾶιξ)/ισαι εἰς/ιν; καὶ αἰ (υπβηθ, ηθδ)/αρα δ/υο)ορθᾶιξ)/ισαι εἰς/ιν.

(η)/αρα εἰς τπαραλλ/ηλουξ ἐυθείαξ ἐυθε=ια)εμπ/ιπτουσα τ/αξ τε)εναλλγων/ιαξ)/ισαξ)αλλ/ηλαιξ ποιε=ι καὶ τ)εκττ=η)εντκαὶ)απεναντ/ιον)/ισην καὶ τ)εντκαὶ)επτα)υτμ/ερη δυσ)ορθᾶιξ)/ισαξ; (/οπερ)εδει δε=ιχαι.

Proposition 29

A straight-line falling across parallel straight-lines makes the alternate angles equal to one another, the external (angle) equal to the internal and opposite (angle), and the (sum of the) internal (angles) on the same side equal to two right-angles.

For let the straight-line EF fall across the parallel straight-lines AB and CD . I say that it makes the alternate angles, AGH and GHD , equal, the external angle EGB equal to the internal and opposite (angle) GHD , and the (sum of the) internal (angles) on the same side, BGH and GHD , equal to two right-angles.

For if AGH is unequal to GHD (then) one of them is greater. Let AGH be greater. Let BGH be added to both. Thus, (the sum of) AGH and BGH is greater than (the sum of) BGH and GHD . But, (the sum of) AGH and BGH is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) BGH and GHD is [also] less than two right-angles. But (straight-lines) being produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, AB and CD , being produced to infinity, will meet together. But they do not meet, on account of them (initially) being assumed parallel (to one another) [Def. 1.23]. Thus, AGH is not unequal to GHD . Thus, (it is) equal. But, AGH is equal to EGB [Prop. 1.15]. And EGB is thus also equal to GHD . Let BGH be added to both. Thus, (the sum of) EGB and BGH is equal to (the sum of) BGH and GHD . But, (the sum of) EGB and BGH is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) BGH and GHD is also equal to two right-angles.

Thus, a straight-line falling across parallel straight-lines makes the alternate angles equal to one another, the external (angle) equal to the internal and opposite (angle), and the (sum of the) internal (angles) on the same side equal to two right-angles. (Which is) the very thing it was required to show.

30.

ἂν τ=η α)υτ=η ἐϋθεία παρ/ἀλληλοι καὶ)ἀλλ/ηλαιξ εἰςπαρ/ἀλληλοι.
)/εστω (εκατ/ερα τ=ων αβ, γδ τ=η εζ παρ/ἀλληλοξ; λ/εγω, (/οτι καὶ (η αβ τ=η γδ
)εστι παρ/ἀλληλοξ.
)εμπιπτ/ετω γείξ α)υτέυθε=ια (η ηκ.

καὶ)επὲι εἰξ παραλλ/ηλουξ ἐϋθείαξ ταβ, εζ ἐυθε=ια)εμπ/επτωκεν (η ηκ,)/ιση)/αρα
 (η (υπαηκ τ=η (υπηθζ. π/αλιν,)επὲι εἰξ παραλλ/ηλουξ ἐϋθείαξ τεζ, γδ ἐυθε=ια)εμπ/επτωκεν
 (η ηκ,)/ιση)εστ(η (υπηθζ τ=η (υπηκδ.)εδείθθη δκαὶ (η (υπαηκ τ=η (υπηθζ)/ιση. καὶ (η
 (υπαηκ)/αρα τ=η (υπηκδ)εστιν)/ιση; καὶ εἰσιν)εναλλ/αχ. παρ/ἀλληλοξ)/αρα)εστ(η
 αβ τ=η γδ.

[ἂν)/αρα τ=η α)υτ=η ἐϋθεία παρ/ἀλληλοι καὶ)ἀλλ/ηλαιξ εἰςπαρ/ἀλληλοι;] (/οπερ
)/εδει δε=ιχαί.

Proposition 30

(Straight-lines) parallel to the same straight-line are also parallel to one another.

Let each of the (straight-lines) AB and CD be parallel to EF . I say that AB is also parallel to CD .

For let the straight-line GK fall across (AB , CD , and EF).

And since the straight-line GK has fallen across the parallel straight-lines AB and EF , (angle) AGK (is) thus equal to GHF [Prop. 1.29]. Again, since the straight-line GK has fallen across the parallel straight-lines EF and CD , (angle) GHF is equal to GKD [Prop. 1.29]. But AGK was also shown (to be) equal to GHF . Thus, AGK is also equal to GKD . And they are alternate (angles). Thus, AB is parallel to CD [Prop. 1.27].

[Thus, (straight-lines) parallel to the same straight-line are also parallel to one another.] (Which is) the very thing it was required to show.

31.

διὰ τοῦ δοθέντος σημείου τῆς δοθείσης ἐνθείας παρ' ἀλλήλων ἐνθεῖαν γραμμὴ ἀγαγεῖν.
)/εστὼ τμδοθσημεῖον τὰ, (ἡ δδοθεῖσα ἐνθεῖα (ἡ βγ; δεῖ δδιὰ τοῦ α σημείου τῆς
 βγ ἐνθείας παρ' ἀλλήλων ἐνθεῖαν γραμμὴ ἀγαγεῖν.
 εἰληφθὼ)επτῆς βγ τυθσημεῖον τδ, καὶ)επεξέυθθω (ἡ αδ; καὶ συνεστ/ατὼ πρτῆς
 δα ἐνθείας καὶ τ=ω πρ(α)υτῆς σημείω τ=ω α τῆς (υπαδγ γων/ιας)/ιση (ἡ (υπδαε; καὶ
)εκβεβλησθὼ)επ) ἐνθείας τῆς εα ἐνθεῖα (ἡ αζ.

καὶ)επὲι εἰς δ/υο ἐνθείας τβγ, εζ ἐνθεῖα)εμπ/ιπτουσα (ἡ αδ τ) ἐναλλγων/ιας τ(υπεαδ,
 αδγ)/ισαξ)αλλ/ηλαιξ πεποίηκεν, παρ' ἀλλήλοξ)/αρα)εστ(ἡ εαζ τῆς βγ.

διὰ τοῦ δοθέντος)/αρα σημείου τοῦ α τῆς δοθείσης ἐνθείας τῆς βγ παρ' ἀλλήλοξ
 ἐνθεῖα γραμμὴ)ηκται (ἡ εαζ; (/οπερ)/εδει ποῖησαι.

Proposition 31

To draw a straight-line parallel to a given straight-line, through a given point.

Let A be the given point, and BC the given straight-line. So it is required to draw a straight-line parallel to the straight-line BC , through the point A .

Let the point D be taken a random on BC , and let AD be joined. And let (angle) DAE , equal to angle ADC , be constructed on the straight-line DA at the point A on it [Prop. 1.23]. And let the straight-line AF be produced in a straight-line with EA .

And since the straight-line AD , (in) falling across the two straight-lines BC and EF , has made the alternate angles EAD and ADC equal to one another, EAF is thus parallel to BC [Prop. 1.27].

Thus, the straight-line EAF has been drawn parallel to the given straight-line BC , through the given point A . (Which is) the very thing it was required to do.

32.

πανττριγ/ωνου μῖαξ τ=ων πλευρ=ων προσεκβληθείσης (ἡ)εκτγων/ια δυσταῖξ)εντκαὶ
)απεναντ/ιον)/ιση)εστ/ιν, καὶ αἱ)ενττο=υ τριγ/ωνου τρεῖς γων/ιαὶ δυσ(ορθαῖξ)/ισαὶ
 εἰς/ιν.

)/εστω τριγωνον ταβγ, καὶ προσεκβεβλ/ησθω α)υτο=υ μ/ια πλευρ(η βγ)επτδ; λ/εγω, (/οτι (η)εκτγων/ια (η (υπαγδ)/ιση)εστδυστᾱιξ)εντκαὶ)απεναντ/ιον τᾱιξ (υπαβ, αβγ, καὶ αὶ)ενττο=υ τριγ/ωνου τρε=ιξ γων/ιαὶ αὶ (υπαβγ, βγα, γαβ δυς)ορθᾱιξ)/ισαὶ εἰς/ιν.

)ηθθω γδία το=υ γ σημείου τ=η αβ εὐθεία παρ/αλληλοξ (η γε.

καὶ)επεί παρ/αλληλ/οξ)εστιν (η αβ τ=η γε, καὶ εἰξ α)υτ)εμπ/επτωκεν (η αγ, αὶ)εναλλγων/ιαὶ αὶ (υπβαγ, αγε)/ισαὶ)αλλ/ηλαιξ εἰς/ιν. π/αλιν,)επεί παρ/αλληλ/οξ)εστιν (η αβ τ=η γε, καὶ εἰξ α)υτ)εμπ/επτωκεν εὐθε=ια (η βδ, (η)εκτγων/ια (η (υπεγδ)/ιση)εσττ=η)εντκαὶ)απεναντ/ιον τ=η (υπαβγ.)εδείθθη δκαὶ (η (υπαγε τ=η (υπβαγ)/ιση; (/ολη)/αρα (η (υπαγδ γων/ια)/ιση)εστδυστᾱιξ)εντκαὶ)απεναντ/ιον τᾱιξ (υπβαγ, αβγ.

κοινπροσκέισθω (η (υπαγβ; αὶ)/αρα (υπαγδ, αβγ τριςτᾱιξ (υπαβγ, βγα, γαβ)/ισαὶ εἰς/ιν.)αλλ) αὶ (υπαγδ, αβγ δυς)ορθᾱιξ)/ισαὶ εἰς/ιν; καὶ αὶ (υπαγβ, γβα, γαβ)/αρα δυς)ορθᾱιξ)/ισαὶ εἰς/ιν.

παντ)/αρα τριγ/ωνου μῖαξ τ=ων πλευρ=ων προσεκβληθείσηξ (η)εκτγων/ια δυςτᾱιξ)εντκαὶ)απεναντ/ιον)/ιση)εστ/ιν, καὶ αὶ)ενττο=υ τριγ/ωνου τρε=ιξ γων/ιαὶ δυς)ορθᾱιξ)/ισαὶ εἰς/ιν; (/οπερ)εδει δε=ιχαί.

Proposition 32

In any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles.

Let ABC be a triangle, and let one of its sides BC be produced to D . I say that the external angle ACD is equal to the (sum of the) two internal and opposite angles CAB and ABC , and the (sum of the) three internal angles of the triangle— ABC , BCA , and CAB —is equal to two right-angles.

For let CE be drawn through point C parallel to the straight-line AB [Prop. 1.31].

And since AB is parallel to CE , and AC has fallen across them, the alternate angles BAC and ACE are equal to one another [Prop. 1.29]. Again, since AB is parallel to CE , and the straight-line BD has fallen across them, the external angle ECD is equal to the internal and opposite (angle) ABC [Prop. 1.29]. But ACE was also shown (to be) equal to BAC . Thus, the whole angle ACD is equal to the (sum of the) two internal and opposite (angles) BAC and ABC .

Let ACB be added to both. Thus, (the sum of) ACD and ACB is equal to the (sum of the) three (angles) ABC , BCA , and CAB . But, (the sum of) ACD and ACB is equal to two right-angles [Prop. 1.13]. Thus, (the sum of) ACB , CBA , and CAB is also equal to two right-angles.

Thus, in any triangle, (if) one of the sides (is) produced (then) the external angle is equal to the (sum of the) two internal and opposite (angles), and the (sum of the) three internal angles of the triangle is equal to two right-angles. (Which is) the very thing it was required to show.

33.

αὶ τ)/ισαξ τε καὶ παραλλ/ηλουξ)επτα)υτμ/ερη)επιζευγν/υουσai εὐθε=ιαὶ καὶ α)υτᾱὶ)/ισαὶ τε καὶ παρ/αλληλοί εἰσιν.

)εστωσαν)/ισαὶ τε καὶ παρ/αλληλοι αὶ αβ, γδ, καὶ)επιζευγν/υωσαν α)υτ)επτα)υτμ/ερη εὐθε=ιαὶ αὶ αγ, βδ; λ/εγω, (/οτι καὶ αὶ αγ, βδ)/ισαὶ τε καὶ παρ/αλληλοί εἰσιν.

)επεξέυθω (η βγ. καὶ)επὲι παρ/αλληλ/οξ)εστιν (η αβ τ=η γδ, καὶ εἰξ α)υτ)εμπ/επτωκεν
 (η βγ, αἱ)εναλλγων/ιαι αἱ (υπαβγ, βγδ)/ισαι)αλλ/ηλαιξ εἰς/ιν. καὶ)επὲι)/ιση)εστ(η
 αβ τ=η γδ κοινδ(η βγ, δ/υο δαἱ αβ, βγ δ/υο τᾶιξ βγ, γδ)/ισαι εἰς/ιν; καὶ γων/ια (η
 (υπαβγ γων/ια τ=η (υπβγδ)/ιση; β/ασιξ)/αρα (η αγ β/ασει τ=η βδ)εστιν)/ιση, καὶ
 ταβγ τρ/ιγωνον τ=ω βγδ τριγ/ωνω)/ισον)εστ/ιν, καὶ αἱ λοιπαἱ γων/ιαι τᾶιξ λοιπᾶιξ
 γων/ιαιξ)/ισαι)/εσονται (εκατ/ερα (εκατ/ερα, (υφ) (αἱ)/ισαι πλευρᾶι (υποτέινουσιν;
)/ιση)/αρα (η (υπαγβ γων/ια τ=η (υπγβδ. καὶ)επὲι εἰξ δ/υο ἐυθείαξ ταγ, βδ ἐυθε=ια
)εμπ/ιπτουσα (η βγ τ)εναλλγων/ιαξ)/ισαξ)αλλ/ηλαιξ πεποίηκεν, παρ/αλληλοξ)/αρα
)εστ(η αγ τ=η βδ.)εδείθθη δα)υτ=η καὶ)/ιση.

αἱ)/αρα τ)/ισαξ τε καὶ παραλλ/ηλουξ)επτα)υτμ/ερη)επιζευγν/ουσαι ἐυθε=ιαι καὶ
 α)υτᾶι)/ισαι τε καὶ παρ/αλληλοί εἰσιν; (/οπερ)/εδει δε=ιχαι.

Proposition 33

Straight-lines joining equal and parallel (straight-lines) on the same sides are themselves also equal and parallel.

Let AB and CD be equal and parallel (straight-lines), and let the straight-lines AC and BD join them on the same sides. I say that AC and BD are also equal and parallel.

Let BC be joined. And since AB is parallel to CD , and BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. And since AB is equal to CD , and BC is common, the two (straight-lines) AB, BC are equal to the two (straight-lines) DC, CB .[†] And the angle ABC is equal to the angle BCD . Thus, the base AC is equal to the base BD , and triangle ABC is equal to triangle DCB [‡], and the remaining angles will be equal to the corresponding remaining angles subtended by the equal sides [Prop. 1.4]. Thus, angle ACB is equal to CBD . Also, since the straight-line BC , (in) falling across the two straight-lines AC and BD , has made the alternate angles (ACB and CBD) equal to one another, AC is thus parallel to BD [Prop. 1.27]. And (AC) was also shown (to be) equal to (BD).

Thus, straight-lines joining equal and parallel (straight-lines) on the same sides are themselves also equal and parallel. (Which is) the very thing it was required to show. [†] The Greek text has “ BC, CD ”, which is obviously a mistake.

[‡] The Greek text has “ DCB ”, which is obviously a mistake.

34.

τ=ων παραλληλογρ/αμμων θωρ/ιων αἱ)απεναντ/ιον πλευράι τε καὶ γων/ιαι)/ισαι)αλλ/ηλαιξ εἰς/ιν, καὶ (η διάμετροξ α)υτδ/ιθα τ/εμνει.

)/εστω παραλληλ/ογραμμον θωρ/ιον ταγδβ, διάμετροξ δα)υτο=υ (η βγ; λ/εγω, (/οτι το=υ αγδβ παραλληλογρ/αμμου αἱ)απεναντ/ιον πλευράι τε καὶ γων/ιαι)/ισαι)αλλ/ηλαιξ εἰς/ιν, καὶ (η βγ διάμετροξ α)υτδ/ιθα τ/εμνει.

)επὲι γπαρ/αλληλ/οξ)εστιν (η αβ τ=η γδ, καὶ εἰξ α)υτ)εμπ/επτωκεν ἐυθε=ια (η βγ, αἱ)εναλλγων/ιαι αἱ (υπαβγ, βγδ)/ισαι)αλλ/ηλαιξ εἰς/ιν. π/αλιν)επὲι παρ/αλληλ/οξ)εστιν (η αγ τ=η βδ, καὶ εἰξ α)υτ)εμπ/επτωκεν (η βγ, αἱ)εναλλγων/ιαι αἱ (υπαγβ, γβδ)/ισαι)αλλ/ηλαιξ εἰς/ιν. δ/υο δτρ/ιγων/α)εστι ταβγ, βγδ τδ/υο γων/ιαξ τ(υπαβγ, βγα δυστᾶιξ (υπβγδ, γβδ)/ισαξ)/εθοντα (εκατ/εραν (εκατ/ερα καὶ μ/ιαν πλευρμῖα πλευρ=α

)/ισην τπρτᾱιξ)/ισαιξ γων/ιαιξ κοινα)υτ=ων τβγ; καὶ τλοιπ)/αρα πλευρτᾱιξ λοιπᾱιξ)/ισαξ (/εχει (εκατ/εραν (εκατ/ερα καὶ τλοιπγων/ιαν τ=η λοιπ=η γων/ια;)/ιση)/αρα (η μαβ πλευρτ=η γδ, (η δαγ τ=η βδ, καὶ)/ετι)/ιση)εστ(η (υπβαγ γων/ια τ=η (υπγδβ. καὶ)επεί)/ιση)εστ(η μ(υπαβγ γων/ια τ=η (υπβγδ, (η δ(υπγβδ τ=η (υπαγβ, (/ολη)/αρα (η (υπαβδ (/ολη τ=η (υπαγδ)εστιν)/ιση.)εδείθθη δκαὶ (η (υπβαγ τ=η (υπγδβ)/ιση.

τ=ων)/αρα παραλληλογρ/αμμων θωρ/ιων ᾱι)απεναντ/ιον πλευράι τε καὶ γων/ιαι)/ισαι)αλλ/ηλαιξ εἰς/ιν.

λ/εγω δ/η, (/οτι καὶ (η διάμετροξ α)υτδ/ιθα τ/εμνει.)επεί γ)/ιση)εστ(η αβ τ=η γδ, κοινδ(η βγ, δ/υο δᾱι αβ, βγ δυστᾱιξ γδ, βγ)/ισαι εἰς(εκατ/ερα (εκατ/ερα; καὶ γων/ια (η (υπαβγ γων/ια τ=η (υπβγδ)/ιση. καὶ β/ασιξ)/αρα (η αγ τ=η δβ)/ιση. καὶ ταβγ []/αρα] τριγωνον τ=ω βγδ τριγ/ωνω)/ισον)εστ/ιν.

(η)/αρα βγ διάμετροξ δ/ιθα τ/εμνει ταβγδ παραλληλ/ογραμμον; (/οπερ)/εδει δε=ιχαι.

Proposition 34

In parallelogramic figures, the opposite sides and angles are equal to one another, and a diagonal cuts them in half.

Let $ACDB$ be a parallelogramic figure, and BC its diagonal. I say that for parallelogram $ACDB$, the opposite sides and angles are equal to one another, and the diagonal BC cuts it in half.

For since AB is parallel to CD , and the straight-line BC has fallen across them, the alternate angles ABC and BCD are equal to one another [Prop. 1.29]. Again, since AC is parallel to BD , and BC has fallen across them, the alternate angles ACB and CBD are equal to one another [Prop. 1.29]. So ABC and BCD are two triangles having the two angles ABC and BCA equal to the two (angles) BCD and CBD , respectively, and one side equal to one side—the (one) by the equal angles and common to them, (namely) BC . Thus, they will also have the remaining sides equal to the corresponding remaining (sides), and the remaining angle (equal) to the remaining angle [Prop. 1.26]. Thus, side AB is equal to CD , and AC to BD . Furthermore, angle BAC is equal to CDB . And since angle ABC is equal to BCD , and CBD to ACB , the whole (angle) ABD is thus equal to the whole (angle) ACD . And BAC was also shown (to be) equal to CDB .

Thus, in parallelogramic figures, the opposite sides and angles are equal to one another.

And, I also say that a diagonal cuts them in half. For since AB is equal to CD , and BC (is) common, the two (straight-lines) AB, BC are equal to the two (straight-lines) DC, CB [†], respectively. And angle ABC is equal to angle BCD . Thus, the base AC (is) also equal to DB , and triangle ABC is equal to triangle BCD [Prop. 1.4].

Thus, the diagonal BC cuts the parallelogram $ACDB$ [‡] in half. (Which is) the very thing it was required to show. [†] The Greek text has “ CD, BC ”, which is obviously a mistake.

[‡] The Greek text has “ $ABCD$ ”, which is obviously a mistake.

35.

τπαραλληλ/ογραμμο τ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα καὶ)εν τᾱιξ α)υτᾱιξ παραλληλοιξ)/ισα)αλλ/ηλοιξ)εστ/ιν.

)εστω παραλληλ/ογραμμο ταβγδ, εβγξ)επτ=ηξ α)υτ=ηξ β/ασεωξ τ=ηξ βγ καὶ)εν τᾱιξ α)υτᾱιξ παραλληλοιξ τᾱιξ αζ, βγ; λ/εγω, (/οτι)/ισον)εστταβγδ τ=ω εβγξ παραλληλογρ/αμμο.

ἔπειτα γπαρὰλληλ/ογραμμ/ον)εστὶ ταβγδ,)/ισὴ)εστ(ἡ ἀδ τ=ἡ βγ. διὰ ταυτὸς καὶ (ἡ ἐξ τ=ἡ βγ)εστὶν)/ισὴ; (/ώστε καὶ (ἡ ἀδ τ=ἡ ἐξ)εστὶν)/ισὴ; καὶ κοιν(ἡ δε; (/ολῇ)/ἀρα (ἡ ἀε (/ολῇ τ=ἡ δζ)εστὶν)/ισὴ.)/εστὶ δὲ καὶ (ἡ ἀβ τ=ἡ δγ)/ισὴ; δ/υο δᾶι εα, ἀβ δ/υο τᾶιξ ζδ, δγ)/ισαὶ εἰς(ἐκατ/ερα (ἐκατ/ερα; καὶ γων/ια (ἡ (υπζδγ γων/ια τ=ἡ (υπεαβ)εστὶν)/ισὴ (ἡ)εκττ=ἡ)εντ/οξ; β/ασιξ)/ἀρα (ἡ ἐβ β/ασει τ=ἡ ζγ)/ισὴ)εστ/ιν, καὶ τεαβ τρ/ιγώνων τ=ω δζγ τριγ/ωνω)/ισον)/εστὰι; κοιν)αφηρ/ησθω τδῃ; λοιπ)/ἀρα ταβηδ τραπ/εζιον λοιπ=ω τ=ω ἐηγζ τραπεζ/ιω)εστ)/ισον; κοινπροσκέισθω τηβγ τρ/ιγώνων; (/ολον)/ἀρα ταβγδ παρὰλληλ/ογραμμον (/ολω τ=ω ἐβγζ παρὰλληλογρ/αμμω)/ισον)εστ/ιν.

τ)/ἀρα παρὰλληλ/ογραμμοι τ)επτ=ἡξ α)υτ=ἡξ β/ασεωξ)/οντα καὶ)εν τᾶιξ α)υτᾶιξ παρὰλληλοῖξ)/ισα)αλλ/ηλοῖξ)εστ/ιν; (/οπερ)/εδεῖ δε=ιχαί.

Proposition 35

Parallelograms which are on the same base and between the same parallels are equal[†] to one another.

Let $ABCD$ and $EBCF$ be parallelograms on the same base BC , and between the same parallels AF and BC . I say that $ABCD$ is equal to parallelogram $EBCF$.

For since $ABCD$ is a parallelogram, AD is equal to BC [Prop. 1.34]. So, for the same (reasons), EF is also equal to BC . So AD is also equal to EF . And DE is common. Thus, the whole (straight-line) AE is equal to the whole (straight-line) DF . And AB is also equal to DC . So the two (straight-lines) EA , AB are equal to the two (straight-lines) FD , DC , respectively. And angle FDC is equal to angle EAB , the external to the internal [Prop. 1.29]. Thus, the base EB is equal to the base FC , and triangle EAB will be equal to triangle DFC [Prop. 1.4]. Let DGE be taken away from both. Thus, the remaining trapezium $ABGD$ is equal to the remaining trapezium $EGCF$. Let triangle GBC be added to both. Thus, the whole parallelogram $ABCD$ is equal to the whole parallelogram $EBCF$.

Thus, parallelograms which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show. [†] Here, for the first time, “equal” means “equal in area”, rather than “congruent”.

36.

τπαρὰλληλ/ογραμμοι τ)επ)/ισων β/ασεων)/οντα καὶ)εν τᾶιξ α)υτᾶιξ παρὰλληλοῖξ)/ισα)αλλ/ηλοῖξ)εστ/ιν.

)εστω παρὰλληλ/ογραμμοι ταβγδ, ἐζηθ)επ)/ισων β/ασεων)/οντα τ=ων βγ, ζη καὶ)εν τᾶιξ α)υτᾶιξ παρὰλληλοῖξ τᾶιξ αθ, βῃ; λ/εγω, (/οτι)/ισον)εστταβγδ παρὰλληλ/ογραμμον τ=ω ἐζηθ.

)επεξέυθθωσαν γᾶι βε, γθ. καὶ)επὶ)/ισὴ)εστ(ἡ βγ τ=ἡ ζη,)αλλ(ἡ ζη τ=ἡ εθ)εστὶν)/ισὴ, καὶ (ἡ βγ)/ἀρα τ=ἡ εθ)εστὶν)/ισὴ. εἰςδὲ καὶ παρ/αλληλοῖ. καὶ)επιζευγν/ουουσιν α)υτᾶι ἐβ, θγ; αἱ δτ)/ισαξ τε καὶ παρὰλληλοῖξ)επτα)υτμ/ερῃ)επιζευγν/ουουσαι)/ισαὶ τε καὶ παρ/αλληλοῖ εἰσι [καὶ αἱ ἐβ, θγ)/ἀρα)/ισαὶ τ/ε εἰσι καὶ παρ/αλληλοῖ]. παρὰλληλ/ογραμμον)/ἀρα)εσττεβγθ. καὶ)εστὶν)/ισον τ=ω ἀβγδ; β/ασιν τε γα)υτ=ω τα)υτ)/εθει τβγ, καὶ)εν τᾶιξ α)υτᾶιξ παρὰλληλοῖξ)εστα)υτ=ω τᾶιξ βγ, αθ. δταυτδὲ καὶ τεζηθ τ=ω α)υτ=ω τ=ω ἐβγθ)εστὶν)/ισον; (/ώστε καὶ ταβγδ παρὰλληλ/ογραμμον τ=ω ἐζηθ)εστὶν)/ισον.

τ)/ἀρα παρὰλληλ/ογραμμοι τ)επ)/ισων β/ασεων)/οντα καὶ)εν τᾶιξ α)υτᾶιξ παρὰλληλοῖξ)/ισα)αλλ/ηλοῖξ)εστ/ιν; (/οπερ)/εδεῖ δε=ιχαί.

Proposition 36

Parallelograms which are on equal bases and between the same parallels are equal to one another.

Let $ABCD$ and $EFGH$ be parallelograms which are on the equal bases BC and FG , and (are) between the same parallels AH and BG . I say that the parallelogram $ABCD$ is equal to $EFGH$.

For let BE and CH be joined. And since BC is equal to FG , but FG is equal to EH [Prop. 1.34], BC is thus equal to EH . And they are also parallel, and EB and HC join them. But (straight-lines) joining equal and parallel (straight-lines) on the same sides are (themselves) equal and parallel [Prop. 1.33] [thus, EB and HC are also equal and parallel]. Thus, $EBCH$ is a parallelogram [Prop. 1.34], and is equal to $ABCD$. For it has the same base, BC , (as $ABCD$), and is between the same parallels, BC and AH , (as $ABCD$) [Prop. 1.35]. So, for the same (reasons), $EFGH$ is also equal to the same (parallelogram) $EBCH$ [Prop. 1.34]. So that the parallelogram $ABCD$ is also equal to $EFGH$.

Thus, parallelograms which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

37.

τρ/ιγωνα τ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)/ισα
αλλ/ηλοιξ)εστ/ιν.

)εστω τρ/ιγωνα ταβγ, δβγ)επτ=ηξ α)υτ=ηξ β/ασεωξ τ=ηξ βγ και)εν τ̃αιξ α)υτ̃αιξ
παραλλ/ηλοιξ τ̃αιξ αδ, βγ; λεγω, (/οτι)/ισον)εστταβγ τρ/ιγωνον τ=ω δβγ τριγ/ωνω.

)εκβεβλ/ησθω (η αδ)εφ) (εκ/ατερα τμ/ερη)επτε, ζ, και δια μτο=υ β τ=η γα παρ/αλληλοξ
)ηθθω (η βε, δδο=υ γ τ=η βδ παρ/αλληλοξ)ηθθω (η γζ. παραλληλ/ογραμμον)/αρα
)εστ(εκ/ατερον τ=ων εβγα, δβγζ; και εισιν)/ισα;)επ/ι τε γτ=ηξ α)υτ=ηξ β/ασέωξ εισι
τ=ηξ βγ και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ τ̃αιξ βγ, εζ; και)εστι το=υ μεβγα παραλληλογρ/αμμου
(/ημισυ ταβγ τρ/ιγωνον; (η γαβ διάμετροξ α)υτδ/ιθα τ/εμνει; το=υ δδβγζ παραλληλογρ/αμμου
(/ημισυ τδβγ τρ/ιγωνον; (η γδγ διάμετροξ α)υτδ/ιθα τ/εμνει. [τδο=ων)/ισων (ημ/ιση
)/ισα)αλλ/ηλοιξ)εστ/ιν].)/ισον)/αρα)εστταβγ τρ/ιγωνον τ=ω δβγ τριγ/ωνω.

τ)/αρα τρ/ιγωνα τ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ
)/ισα)αλλ/ηλοιξ)εστ/ιν; (/οπερ)εδει δε=ιχαι.

Proposition 37

Triangles which are on the same base and between the same parallels are equal to one another.

Let ABC and DBC be triangles on the same base BC , and between the same parallels AD and BC . I say that triangle ABC is equal to triangle DBC .

Let AD be produced in both directions to E and F , and let the (straight-line) BE be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) CF be drawn through C parallel to BD [Prop. 1.31]. Thus, $EBCA$ and $DBC F$ are both parallelograms, and are equal. For they are on the same base BC , and between the same parallels BC and EF [Prop. 1.35]. And the triangle ABC is half of the parallelogram $EBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And

the triangle DBC (is) half of the parallelogram $DBCF$. For the diagonal DC cuts the latter in half [Prop. 1.34]. [And the halves of equal things are equal to one another.][†] Thus, triangle ABC is equal to triangle DBC .

Thus, triangles which are on the same base and between the same parallels are equal to one another. (Which is) the very thing it was required to show. [†] This is an additional common notion.

38.

ττρ/ιγωνα τ)επ)/ισων β/ασεων)/οντα καὶ)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ)/ισα)αλλ/ηλοιξ)εστ/ιν.

)εστω τρ/ιγωνα ταβγ, δεξ)επ)/ισων β/ασεων τ=ων βγ, εζ καὶ)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ τᾱιξ βζ, αδ; λ/εγω, (/οτι)/ισον)εστταβγ τρ/ιγωνον τ=ω δεξ τριγ/ωνω.

)εκβεβλ/ησθω γ(η αδ)εφ) (εκ/ατερα τμ/ερη)επτη, θ, καὶ διὰ μτο=υ β τ=η γα παρ/αλληλοξ)/ηθθω (η βη, δδο=υ ζ τ=η δε παρ/αλληλοξ)/ηθθω (η ζθ. παραλληλ/ογραμμον)/αρα)εστ(εκ/ατερον τ=ων ηβγα, δεξθ; καὶ)/ισον τηβγα τ=ω δεξθ;)επ/ι τε γ)/ισων β/ασέων)ίσι τ=ων βγ, εζ καὶ)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ τᾱιξ βζ, ηθ; κάι)εστι το=υ μηβγα παραλληλογρ/αμμου (/ημισυ ταβγ τρ/ιγωνον. (η γαβ διάμετροξ α)υτδ/ιθα τ/εμνει; το=υ δδεξθ παραλληλογρ/αμμου (/ημισυ τζεδ τρ/ιγωνον; (η γδζ δ/ιαμετροξ α)υτδ/ιθα τ/εμνει [τδτ=ων)/ισων (ημ/ιση)/ισα)αλλ/ηλοιξ)εστ/ιν].)/ισον)/αρα)εστταβγ τρ/ιγωνον τ=ω δεξ τριγ/ωνω.

τ)/αρα τρ/ιγωνα τ)επ)/ισων β/ασεων)/οντα καὶ)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ)/ισα)αλλ/ηλοιξ)εστ/ιν; (/οπερ)/εδει δε=ιχαί.

Proposition 38

Triangles which are on equal bases and between the same parallels are equal to one another.

Let ABC and DEF be triangles on the equal bases BC and EF , and between the same parallels BF and AD . I say that triangle ABC is equal to triangle DEF .

For let AD be produced in both directions to G and H , and let the (straight-line) BG be drawn through B parallel to CA [Prop. 1.31], and let the (straight-line) FH be drawn through F parallel to DE [Prop. 1.31]. Thus, $GBCA$ and $DEFH$ are each parallelograms. And $GBCA$ is equal to $DEFH$. For they are on the equal bases BC and EF , and between the same parallels BF and GH [Prop. 1.36]. And triangle ABC is half of the parallelogram $GBCA$. For the diagonal AB cuts the latter in half [Prop. 1.34]. And triangle FED (is) half of parallelogram $DEFH$. For the diagonal DF cuts the latter in half. [And the halves of equal things are equal to one another.] Thus, triangle ABC is equal to triangle DEF .

Thus, triangles which are on equal bases and between the same parallels are equal to one another. (Which is) the very thing it was required to show.

39.

τ)/ισα τριγωνα τ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα και)επτα)υτμ/ερη και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν.

)εστω)/ισα τριγωνα ταβγ, δβγ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα και)επτα)υτμ/ερη τ=ηξ βγ; λεγω, (/οτι και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν.

)επεζέυθω γ(η αδ; λεγω, (/οτι παρ/αλληλ/οξ)εστιν (η αδ τ=η βγ.

εί γμ/η,)/ηθθω δια το=υ α σημείου τ=η βγ εύθεία παρ/αλληλοξ (η αε, και)επεζέυθω (η εγ.)/ισον)/αρα)εστταβγ τριγωνον τ=ω εβγ τριγ/ωνω;)επ/ι τε γτ=ηξ α)υτ=ηξ β/ασέωξ)εστιν α)υτ=ω τ=ηξ βγ και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ.)αλλαταβγ τ=ω δβγ)εστιν)/ισον; και τδβγ)/αρα τ=ω εβγ)/ισον)εσττμε=ιζον τ=ω)ελ/ασσονι; (/οπερ)εστ)αδ/υνατον; ουκ)/αρα παρ/αλληλ/οξ)εστιν (η αε τ=η βγ. (ομοίωξ δδείχομεν, (/οτι ουδ))/αλλη τιξ πλτ=ηξ αδ; (η αδ)/αρα τ=η βγ)εστι παρ/αλληλοξ.

τ)/αρα)/ισα τριγωνα τ)επτ=ηξ α)υτ=ηξ β/ασεωξ)/οντα και)επτα)υτμ/ερη και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν; (/οπερ)/εδει δε=ιχαι.

Proposition 39

Equal triangles which are on the same base, and on the same side, are also between the same parallels.

Let ABC and DBC be equal triangles which are on the same base BC , and on the same side (of it). I say that they are also between the same parallels.

For let AD be joined. I say that AD and BC are parallel.

For, if not, let AE be drawn through point A parallel to the straight-line BC [Prop. 1.31], and let EC be joined. Thus, triangle ABC is equal to triangle EBC . For it is on the same base as it, BC , and between the same parallels [Prop. 1.37]. But ABC is equal to DBC . Thus, DBC is also equal to EBC , the greater to the lesser. The very thing is impossible. Thus, AE is not parallel to BC . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BC .

Thus, equal triangles which are on the same base, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show.

40.

τ)/ισα τριγωνα τ)επ)/ισων β/ασεων)/οντα και)επτα)υτμ/ερη και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν.

)εστω)/ισα τριγωνα ταβγ, γδε)επ)/ισων β/ασεων τ=ων βγ, γε και)επτα)υτμ/ερη. λεγω, (/οτι και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν.

)επεζέυθω γ(η αδ; λεγω, (/οτι παρ/αλληλ/οξ)εστιν (η αδ τ=η βε.

εί γμ/η,)/ηθθω δια το=υ α τ=η βε παρ/αλληλοξ (η αζ, και)επεζέυθω (η ζε.)/ισον)/αρα)εστταβγ τριγωνον τ=ω ζγε τριγ/ωνω;)επ/ι τε γ)/ισων β/ασέων είσι τ=ων βγ, γε και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ τ̃αιξ βε, αζ.)αλλαταβγ τριγωνον)/ισον)εσττ=ω δγε [τριγ/ωνω]; και τδγε)/αρα [τριγ/ωνον])/ισον)εσττ=ω ζγε τριγ/ωνω τμε=ιζον τ=ω)ελ/ασσονι; (/οπερ)εστ)αδ/υνατον; ουκ)/αρα παρ/αλληλοξ (η αζ τ=η βε. (ομοίωξ δδείχομεν, (/οτι ουδ))/αλλη τιξ πλτ=ηξ αδ; (η αδ)/αρα τ=η βε)εστι παρ/αλληλοξ.

τ)/αρα)/ισα τριγωνα τ)επ)/ισων β/ασεων)/οντα και)επτα)υτμ/ερη και)εν τ̃αιξ α)υτ̃αιξ παραλλ/ηλοιξ)εστ/ιν; (/οπερ)/εδει δε=ιχαι.

Proposition 40[†]

Equal triangles which are on equal bases, and on the same side, are also between the same parallels.

Let ABC and CDE be equal triangles on the equal bases BC and CE (respectively), and on the same side (of BE). I say that they are also between the same parallels.

For let AD be joined. I say that AD is parallel to BE .

For if not, let AF be drawn through A parallel to BE [Prop. 1.31], and let FE be joined. Thus, triangle ABC is equal to triangle FCE . For they are on equal bases, BC and CE , and between the same parallels, BE and AF [Prop. 1.38]. But, triangle ABC is equal to [triangle] DCE . Thus, [triangle] DCE is also equal to triangle FCE , the greater to the lesser. The very thing is impossible. Thus, AF is not parallel to BE . Similarly, we can show that neither (is) any other (straight-line) than AD . Thus, AD is parallel to BE .

Thus, equal triangles which are on equal bases, and on the same side, are also between the same parallels. (Which is) the very thing it was required to show. [†] This whole proposition is regarded by Heiberg as a relatively early interpolation to the original text.

41.

)εαν παραλληλ/ογραμμον τριγ/ωνω β/ασιν τε)/εθ η τα)υτ καί)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ)=η, διπλ/ασίον)εστ/ι τπαραλληλ/ογραμμον το=υ τριγ/ωνου.

παραλληλ/ογραμμον γταβγδ τριγ/ωνω τ=ω εβγ β/ασιν τε)εθ/ετω τα)υτ βγ καί)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ)/εστω τᾱιξ βγ, αε; λ/εγω, (/οτι διπλ/ασίον)εστι ταβγδ παραλληλ/ογραμμον το=υ βεγ τριγ/ωνου.

)επεξέυθθω γ(η αγ.)/ισον δ/η)εστι ταβγ τριγ/ωνων τ=ω εβγ τριγ/ωνω;)επ/ι τε γτ=ηξ α)υτ=ηξ β/ασέωξ)εστιν α)υτ=ω τ=ηξ βγ καί)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ τᾱιξ βγ, αε.)αλλα ταβγδ παραλληλ/ογραμμον διπλ/ασίον)εστι το=υ αβγ τριγ/ωνου; (η γαγ διάμετροξ α)υτδ/ιθα τ/εμνει; (/ωστε ταβγδ παραλληλ/ογραμμον καί το=υ εβγ τριγ/ωνου)εστδιπλ/ασιον.

)εαν)/αρα παραλληλ/ογραμμον τριγ/ωνω β/ασιν τε)/εθ η τα)υτ καί)εν τᾱιξ α)υτᾱιξ παραλλ/ηλοιξ)=η, διπλ/ασίον)εστ/ι τπαραλληλ/ογραμμον το=υ τριγ/ωνου; (/οπερ)/εδει δε=ιχαί.

Proposition 41

If a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle.

For let parallelogram $ABCD$ have the same base BC as triangle EBC , and let it be between the same parallels, BC and AE . I say that parallelogram $ABCD$ is double (the area) of triangle BEC .

For let AC be joined. So triangle ABC is equal to triangle EBC . For it is on the same base, BC , (as EBC), and between the same parallels, BC and AE [Prop. 1.37]. But, parallelogram $ABCD$ is double (the area) of triangle ABC . For the diagonal AC cuts the former in half [Prop. 1.34]. So parallelogram $ABCD$ is also double (the area) of triangle EBC .

Thus, if a parallelogram has the same base as a triangle, and is between the same parallels, (then) the parallelogram is double (the area) of the triangle. (Which is) the very thing it was required to show.

42.

τ=ω δοθ/εντι τριγ/ωνω)/ισον παραλληλ/ογραμμον συστ/ησασθαι)εν τ=η δοθείση γων/ια ἔυθυγρ/αμμω.

)εστω τμδοθτρ/ιγωνον ταβγ, (η δδοθε=ισα γων/ια ἔυθ/υγραμμοξ (η δ; δε=ι δτ=ω αβγ τριγ/ωνω)/ισον παραλληλ/ογραμμον συστ/ησασθαι)εν τ=η δ γων/ια ἔυθυγρ/αμμω.

τετμ/ηοθω (η βγ δ/ιθα καττε, και)επεξέυθθω (η αε, και συνεστ/ατω πρτ=η εγ ἔυθεία και τ=ω πρ)υτ=η σημείω τ=ω ε τ=η δ γων/ια)/ιση (η (υπεξ, και δια μτο=υ α τ=η εγ παρ/αλληλοξ)/ηθθω (η αη, δια δτο=υ γ τ=η εζ παρ/αλληλοξ)/ηθθω (η γη; παραλληλ/ογραμμον)/αρα)εστιζεγη. και)επεί)/ιση)εστ(η βε τ=η εγ,)/ισον)εστκαι ταβε τρ/ιγωνον τ=ω αεγ τριγ/ωνω;)επ/ι τε γ)/ισων β/ασέων είσι τ=ων βε, εγ και)εν τ̂αιξ α)υτ̂αιξ παραλλ/ηλοιξ τ̂αιξ βγ, αη; διπλ/ασιον)/αρα)εστταβγ τρ/ιγωνον το=υ αεγ τριγ/ωνον.)/εστι δκαι τζεγη παραλληλ/ογραμμον διπλ/ασιον το=υ αεγ τριγ/ωνον; β/ασιν τε γ)υτ=ω τα)υτ)/εθει και)εν τ̂αιξ α)υτ̂αιξ)εστιν α)υτ=ω παραλλ/ηλοιξ;)/ισον)/αρα)εστιζεγη παραλληλ/ογραμμον τ=ω αβγ τριγ/ωνω. και)/εθει τ(υπεξ γων/ιαν)/ισην τ=η δοθείση τ=η δ.

τ=ω)/αρα δοθ/εντι τριγ/ωνω τ=ω αβγ)/ισον παραλληλ/ογραμμον συν/εσταται τζεγη)εν γων/ια τ=η (υπεξ, (/ητιξ)εστ)/ιση τ=η δ; (/οπερ)εδει ποιῆσαι.

Proposition 42

To construct a parallelogram equal to a given triangle in a given rectilinear angle.

Let ABC be the given triangle, and D the given rectilinear angle. So it is required to construct a parallelogram equal to triangle ABC in the rectilinear angle D .

Let BC be cut in half at E [Prop. 1.10], and let AE be joined. And let (angle) CEF , equal to angle D , be constructed at the point E on the straight-line EC [Prop. 1.23]. And let AG be drawn through A parallel to EC [Prop. 1.31], and let CG be drawn through C parallel to EF [Prop. 1.31]. Thus, $FECG$ is a parallelogram. And since BE is equal to EC , triangle ABE is also equal to triangle AEC . For they are on the equal bases, BE and EC , and between the same parallels, BC and AG [Prop. 1.38]. Thus, triangle ABC is double (the area) of triangle AEC . And parallelogram $FECG$ is also double (the area) of triangle AEC . For it has the same base (as AEC), and is between the same parallels (as AEC) [Prop. 1.41]. Thus, parallelogram $FECG$ is equal to triangle ABC . ($FECG$) also has the angle CEF equal to the given (angle) D .

Thus, parallelogram $FECG$, equal to the given triangle ABC , has been constructed in the angle CEF , which is equal to D . (Which is) the very thing it was required to do.

43.

παντ παραλληλογρ/αμμου τ=ων περτ διάμετρον παραλληλογρ/αμμων τ παραπληρ/ωματα)/ισα)αλλ/ηλοιξ)εστ/ιν.

)εστω παραλληλ/ογραμμον ταβγδ, διάμετροξ δα)υτο=υ (η αγ, περδταγ παραλληλ/ογραμμοξ)/εστω τεθ, ζη, τδλεγ/ομενα παραπληρ/ωματα τβκ, κδ; λ/εγω, (/οτι)/ισον)εσττβκ παραπληρ/ωμα τ=ω κδ παραπληρ/ωματι.

)επεί γ παραλληλ/ογραμμ/ον)εστι ταβγδ, διάμετροξ δα)υτο=υ (η αγ,)/ισον)εστταβγ τρ/ιγωνον τ=ω αγδ τριγ/ωνω. π/αλιν,)επεί παραλληλ/ογραμμ/ον)εστι τεθ, διάμετροξ δα)υτο=υ)εστιν (η ακ,)/ισον)εστταεκ τρ/ιγωνον τ=ω αθκ τριγ/ωνω. δια τα)υτδκαί τκζγ τρ/ιγωνον τ=ω κηγ)εστιν)/ισον.)επεί δ=υν τμαεκ τρ/ιγωνον τ=ω αθκ τριγ/ωνω)εστ)/ισον, τδκζγ τ=ω κηγ, ταεκ τρ/ιγωνον μεττο=υ κηγ)/ισον)εσττ=ω αθκ τριγ/ωνω μεττο=υ κζγ;)/εστι δκαί (/ολον ταβγ τρ/ιγωνον (/ολω τ=ω αδγ)/ισον; λοιπ)/αρα τβκ παραπληρ/ωμα λοιπ=ω τ=ω κδ παραπληρ/ωματ/ι)εστιν)/ισον.

παντ)/αρα παραλληλογρ/αμμου θωρ/ιου τ=ων περτ διάμετρον παραλληλογρ/αμμων τ παραπληρ/ωματα)/ισα)αλλ/ηλοιξ)εστ/ιν; (/οπερ)/εδει δε=ιχαί.

Proposition 43

For any parallelogram, the complements of the parallelograms about the diagonal are equal to one another.

Let $ABCD$ be a parallelogram, and AC its diagonal. And let EH and FG be the parallelograms about AC , and BK and KD the so-called complements (about AC). I say that the complement BK is equal to the complement KD .

For since $ABCD$ is a parallelogram, and AC its diagonal, triangle ABC is equal to triangle ACD [Prop. 1.34]. Again, since EH is a parallelogram, and AK is its diagonal, triangle AEK is equal to triangle AHK [Prop. 1.34]. So, for the same (reasons), triangle KFC is also equal to (triangle) KGC . Therefore, since triangle AEK is equal to triangle AHK , and KFC to KGC , triangle AEK plus KGC is equal to triangle AHK plus KFC . And the whole triangle ABC is also equal to the whole (triangle) ADC . Thus, the remaining complement BK is equal to the remaining complement KD .

Thus, for any parallelogramic figure, the complements of the parallelograms about the diagonal are equal to one another. (Which is) the very thing it was required to show.

44.

παρτδοθε=ισαν εϋθε=ιαν τ=ω δοθ/εντι τριγ/ωνω)/ισον παραλληλ/ογραμμον παραβαλε=ιν)εν τ=η δοθείση γων/ια εϋθυγρ/αμμοξ.

)εστω (η μδοθε=ισα εϋθε=ια (η αβ, τδδοθτρ/ιγωνον τγ, (η δδοθε=ισα γων/ια εϋθ/υγραμμοξ (η δ; δε=ι δπαρτδοθε=ισαν εϋθε=ιαν ταβ τ=ω δοθ/εντι τριγ/ωνω τ=ω γ)/ισον παραλληλ/ογραμμον παραβαλε=ιν)εν)/ιση τ=η δ γων/ια.

συνεστ/ατω τ=ω γ τριγ/ωνω)/ισον παραλληλ/ογραμμον τβεζη)εν γων/ια τ=η (υπεβη, (η)εστιν)/ιση τ=η δ; και κείσθω (/ωστε)επ) ευθείαξ έ=ιναι τβε τ=η αβ, και διηθθω (η ζη)επτθ, και δια το=υ α (οποτ/ερα τ=ων βη, εζ παρ/αλληλοξ)/ηθθω (η αθ, και)επεζέυθθω (η θβ. και)επεί έιξ παραλλ/ηλουξ ταθ, εζ ευθε=ια)εν/επεσεν (η θζ, άι)/αρα (υπαθζ, θζε γων/ιαι δυσ)ορθάιξ έισιν)/ισαι. άι)/αρα (υπβθη, ηζε δ/υο)ορθ=ων)ελ/ασσον/εξ έισιν; άι δ)απ)ελασς/ονων)δ/υο)ορθ=ων έιξ)/απειρον)εκβαλλ/ομεναι συμπι/πτουσιν; άι θβ, ζε)/αρα)εκβαλλ/ομεναι συμπεσο=υνται.)εκβεβλ/ησθωσαν και συμπιπτ/ετωσαν καττκ, και δια το=υ κ σημείου (οποτ/ερα τ=ων εα, ζθ παρ/αλληλοξ)/ηθθω (η κλ, και)εκβεβλ/ησθωσαν άι θα, ηβ)επτλ, μ σημει=ια. παραλληλ/ογραμμον)/αρα)εστθλκζ, διάμετροξ δα)υτο=υ (η θκ, περδθκ παραλληλ/ογραμματα, με, τδλεγ/ομενα παραπληρ/ωματα τλβ, βζ;)/ισον)/αρα)εσττλβ τ=ω βζ.)αλλτβζ τ=ω γ τριγ/ωνω)εστ)/ισον; και τλβ)/αρα τ=ω γ)εστιν)/ισον. και)επεί)/ιση)εστ(η (υπηβε γων/ια τ=η (υπαβμ,)αλλ(η (υπηβε τ=η δ)εστιν)/ιση, και (η (υπαβμ)/αρα τ=η δ γων/ια)εστ)/ιση. παρτδοθε=ισαν)/αρα ευθε=ιαν ταβ τ=ω δοθ/εντι τριγ/ωνω τ=ω γ)/ισον παραλληλ/ογραμμον παραβ/εβληται τλβ)εν γων/ια τ=η (υπαβμ, (/η)εστιν)/ιση τ=η δ; (/οπερ)/εδει ποιήσαι.

Proposition 44

To apply a parallelogram equal to a given triangle to a given straight-line in a given rectilinear angle.

Let AB be the given straight-line, C the given triangle, and D the given rectilinear angle. So it is required to apply a parallelogram equal to the given triangle C to the given straight-line AB in an angle equal to (angle) D .

Let the parallelogram $BEFG$, equal to the triangle C , be constructed in the angle EBG , which is equal to D [Prop. 1.42]. And let it be placed so that BE is straight-on to AB .[†] And let FG be drawn through to H , and let AH be drawn through A parallel to either of BG or EF [Prop. 1.31], and let HB be joined. And since the straight-line HF falls across the parallels AH and EF , the (sum of the) angles AHF and HFE is thus equal to two right-angles [Prop. 1.29]. Thus, (the sum of) BHG and GFE is less than two right-angles. And (straight-lines) produced to infinity from (internal angles whose sum is) less than two right-angles meet together [Post. 5]. Thus, being produced, HB and FE will meet together. Let them be produced, and let them meet together at K . And let KL be drawn through point K parallel to either of EA or FH [Prop. 1.31]. And let HA and GB be produced to points L and M (respectively). Thus, $HLKF$ is a parallelogram, and HK its diagonal. And AG and ME (are) parallelograms, and LB and BF the so-called complements, about HK . Thus, LB is equal to BF [Prop. 1.43]. But, BF is equal to triangle C . Thus, LB is also equal to C . Also, since angle GBE is equal to ABM [Prop. 1.15], but GBE is equal to D , ABM is thus also equal to angle D .

Thus, the parallelogram LB , equal to the given triangle C , has been applied to the given straight-line AB in the angle ABM , which is equal to D . (Which is) the very thing it was required to do. [†] This can be achieved using Props. 1.3, 1.23, and 1.31.

45.

τ=ω δοθ/εντι ευθυγρ/αμμω)/ισον παραλληλ/ογραμμον συστ/ησασθαι)εν τ=η δοθείση γων/ια ευθυγρ/αμμω.

)εστω τμδοθεύθ/υγραμμον ταβγδ, (η δδοθε=ισα γων/ια ευθ/υγραμμοξ (η ε; δε=ι δτ=ω αβγδ ευθυγρ/αμμω)/ισον παραλληλ/ογραμμον συστ/ησασθαι)εν τ=η δοθείση γων/ια τ=η ε.

)επεξέυθθω (η δβ, καὶ συνεστ/ατω τ=ω αβδ τριγ/ωνω)/ισον παραλληλ/ογραμμον
 τζθ)εν τ=η (υπθκζ γων/ια, (/η)εστιν)/ιση τ=η ε; καὶ παραβεβλ/ησθω παρτηθ έυθε=ιαν
 τ=ω δβγ τριγ/ωνω)/ισον παραλληλ/ογραμμον τημ)εν τ=η (υπηθμ γων/ια, (/η)εστιν
)/ιση τ=η ε. καὶ)επεί (η ε γων/ια (εκατ/ερα τ=ων (υπθκζ, ηθμ)εστιν)/ιση, καὶ (η (υπθκζ
)/αρα τ=η (υπηθμ)εστιν)/ιση. κοινπροσκέισθω (η (υπκθη; άι)/αρα (υπζκθ, κθη τάιξ
 (υπκθη, ηθμ)/ισαι έις/ιν.)αλλ) άι (υπζκθ, κθη δυσ)ορθάιξ)/ισαι έις/ιν; καὶ άι (υπκθη,
 ηθμ)/αρα δ/υο)ορθάιξ)/ισαι έις/ιν. πρδ/η τινι έυθε=ια τ=η ηθ καὶ τ=ω πρα)υτ=η σημείω
 τ=ω θ δ/υο έυθε=ιαι άι κθ, θμ μ)επτα)υτμ/ερη κείμεναι τ)εφεχ=ηξ γων/ιαξ δ/υο)ορθάιξ
)/ισαξ ποιο=ουσιν;)επ) έυθείαξ)/αρα)εστ(η κθ τ=η θμ; καὶ)επεί έίξ παραλλ/ηλουξ
 τκμ, ζη έυθε=ια)εν/επεσεν (η θη, άι)εναλλγων/ιαι άι (υπμθη, θηζ)/ισαι)αλλ/ηλαιξ
 έις/ιν. κοινπροσκέισθω (η (υπθηλ; άι)/αρα (υπμθη, θηλ τάιξ (υπθηζ, θηλ)/ισαι έισιν.
)αλλ) άι (υπμθη, θηλ δ/υο)ορθάιξ)/ισαι έις/ιν; καὶ άι (υπθηζ, θηλ)/αρα δ/υο)ορθάιξ
)/ισαι έις/ιν;)επ) έυθείαξ)/αρα)εστ(η ζη τ=η ηλ. καὶ)επεί (η ζκ τ=η θη)/ιση τε καὶ
 παρ/αλληλ/οξ)εστιν,)αλλκαὶ (η θη τ=η μλ, καὶ (η κζ)/αρα τ=η μλ)/ιση τε καὶ παρ/αλληλ/οξ
)εστιν; καὶ)επιζευγν/ουουσιν α)υτέυθε=ιαι άι κμ, ζλ; καὶ άι κμ, ζλ)/αρα)/ισαι τε καὶ
 παρ/αλληλοί έισιν; παραλληλ/ογραμμον)/αρα)εσττκζλμ. καὶ)επεί)/ισον)εσττμαβδ
 τρ/ιγωνον τ=ω ζθ παραλληλογρ/αμμω, τδδβγ τ=ω ημ, (/ολον)/αρα ταβγδ έυθ/υγραμμον
 (/ολω τ=ω κζλμ παραλληλογρ/αμμω)εστ)/ισον.

τ=ω)/αρα δοθ/εντι έυθυγρ/αμμω τ=ω αβγδ)/ισον παραλληλ/ογραμμον συν/εσταται
 τκζλμ)εν γων/ια τ=η (υπζκμ, (/η)εστιν)/ιση τ=η δοθείση τ=η ε; (/οπερ)/εδει ποιήσαι.

Proposition 45

To construct a parallelogram equal to a given rectilinear figure in a given rectilinear angle.

Let $ABCD$ be the given rectilinear figure,[†] and E the given rectilinear angle. So it is required to construct a parallelogram equal to the rectilinear figure $ABCD$ in the given angle E .

Let DB be joined, and let the parallelogram FH , equal to the triangle ABD , be constructed in the angle HKF , which is equal to E [Prop. 1.42]. And let the parallelogram GM , equal to the triangle DBC , be applied to the straight-line GH in the angle GHM , which is equal to E [Prop. 1.44]. And since angle E is equal to each of (angles) HKF and GHM , (angle) HKF is thus also equal to GHM . Let KHG be added to both. Thus, (the sum of) FKH and KHG is equal to (the sum of) KHG and GHM . But, (the sum of) FKH and KHG is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) KHG and GHM is also equal to two right-angles. So two straight-lines, KH and HM , not lying on the same side, make adjacent angles with some straight-line GH , at the point H on it, (whose sum is) equal to two right-angles. Thus, KH is straight-on to HM [Prop. 1.14]. And since the straight-line HG falls across the parallels KM and FG , the alternate angles mhG and HGF are equal to one another [Prop. 1.29]. Let HGL have been added to both. Thus, (the sum of) mhG and HGL is equal to (the sum of) HGF and HGL . But, (the sum of) mhG and HGL is equal to two right-angles [Prop. 1.29]. Thus, (the sum of) HGF and HGL is also equal to two right-angles. Thus, FG is straight-on to GL [Prop. 1.14]. And since FK is equal and parallel to HG [Prop. 1.34], but also HG to ML [Prop. 1.34], KF is thus also equal and parallel to ML [Prop. 1.30]. And the straight-lines KM and FL join them. Thus, KM and FL are equal and parallel as well [Prop. 1.33]. Thus, $KFLM$ is a parallelogram. And since triangle ABD is equal to parallelogram FH , and DBC to GM , the whole rectilinear figure $ABCD$ is thus equal to the whole parallelogram $KFLM$.

Thus, the parallelogram $KFLM$, equal to the given rectilinear figure $ABCD$, has been constructed in the angle FKM , which is equal to the given (angle) E . (Which is) the very thing it was required to do. [†] The proof is only given for a four-sided figure. However, the extension to many-sided figures is trivial.

46.

)απτ=ηξ δοθείσῃ ξ' ἐυθείᾳ ξ τετρ/αγωνον)αναγρ/αψαι.

) /εστω (η δοθε=ισα ἐυθε=ια (η αβ; δε=ι δ)απτ=ηξ αβ ἐυθείᾳ ξ τετρ/αγωνον)αναγρ/αψαι.

) /ηθθω τ=η αβ ἐυθείᾳ)απτο=υ πρ(α)υτ=η σημείου το=υ α πρ(ο)ρθ(η αγ, καὶ κείσθω τ=η αβ) /ιση (η αδ; καὶ διὰ μτο=υ δ σημείου τ=η αβ παρ/αλληλοξ) /ηθθω (η δε, διὰ δτο=υ β σημείου τ=η αδ παρ/αλληλοξ) /ηθθω (η βε. παραλληλ/ογραμμον) /αρα)εστταδεβ;) /ιση) /αρα)εστ(η μαβ τ=η δε, (η δαδ τ=η βε.)αλλ(η αβ τ=η αδ)εστιν) /ιση; αἱ τ/εσσαρεξ) /αρα αἱ βα, αδ, δε, εβ) /ισαι)αλλ/ηλαιξ εἰς/ιν;)ις/οπλευρον) /αρα)εστταδεβ παραλληλ/ογραμμον. λέγω δ/η, (/οτι καὶ)ορθογ/ωνιον.)επὲι γέιξ παραλλ/ηλουξ ταβ, δε ἐυθε=ια)εν/επεσεν (η αδ, αἱ) /αρα (υπβαδ, αδε γων/ιαι δ/υο)ορθᾶιξ) /ισαι εἰς/ιν.)ορθδ(η (υπβαδ;)ορθ) /αρα καὶ (η (υπαδε. τ=ων δπαραλληλογρ/αμμων θωρ/ιων αἱ)απεναντ/ιον πλευράι τε καὶ γων/ιαι) /ισαι)αλλ/ηλαιξ εἰς/ιν;)ορθ) /αρα καὶ (εκατ/ερα τ=ων)απεναντ/ιον τ=ων (υπαβε, βεδ γωνῶν;)ορθογ/ωνιον) /αρα)εστταδεβ.)εδείθθη δκαὶ)ις/οπλευρον.

τετρ/αγωνον) /αρα)εστ/ιν; καὶ)εστιν)απτ=ηξ αβ ἐυθείᾳ ξ)αναγεγραμμ/ενον; (/οπερ) /εδει ποιῆσαι.

Proposition 46

To describe a square on a given straight-line.

Let AB be the given straight-line. So it is required to describe a square on the straight-line AB .

Let AC be drawn at right-angles to the straight-line AB from the point A on it [Prop. 1.11], and let AD be made equal to AB [Prop. 1.3]. And let DE be drawn through point D parallel to AB [Prop. 1.31], and let BE be drawn through point B parallel to AD [Prop. 1.31]. Thus, $ADEB$ is a parallelogram. Therefore, AB is equal to DE , and AD to BE [Prop. 1.34]. But, AB is equal to AD . Thus, the four (sides) BA , AD , DE , and EB are equal to one another. Thus, the parallelogram $ADEB$ is equilateral. So I say that (it is) also right-angled. For since the straight-line AD falls across the parallels AB and DE , the (sum of the) angles BAD and ADE is equal to two right-angles [Prop. 1.29]. But BAD (is a) right-angle. Thus, ADE (is) also a right-angle. And for parallelogramic figures, the opposite sides and angles are equal to one another [Prop. 1.34]. Thus, each of the opposite angles ABE and BED (are) also right-angles. Thus, $ADEB$ is right-angled. And it was also shown (to be) equilateral.

Thus, ($ADEB$) is a square [Def. 1.22]. And it is described on the straight-line AB . (Which is) the very thing it was required to do.

47.

)εν το=ιξ)ορθογων/ιοιξ τριγ/ωνοιξ τ)απτ=ηξ τ)ορθγων/ιαν (υποτεινούσῃ πλευρ=αξ τετρ/αγωνον) /ισον)εσττο=ιξ)απτ=ων τ)ορθγων/ιαν περιεθους=ων πλευρ=ων τετραγ/ωνοιξ.) /εστω τρ/ιγωνον)ορθογ/ωνιον ταβγ)ορθ) /εθον τ(υπβαγ γων/ιαν; λέγω, (/οτι τ)απτ=ηξ βγ τετρ/αγωνον) /ισον)εσττο=ιξ)απτ=ων βα, αγ τετραγ/ωνοιξ.

)αναγεγρα/αφθω γ)απμτ=ηξ βγ τετρ/αγωνον τβδεγ,)απδτ=ων βα, αγ τηβ, θγ, και
 δια το=υ α (οποτ/ερα τ=ων βδ, γε παρ/αλληλοξ)/ηθθω (η αλ; και)επεξέυθθωσαν
 αι αδ, ζγ. και)επει)ορθ/η)εστιν (εκατ/ερα τ=ων (υπβαγ, βαη γωνίων, πρδ/η τινι
 ευθεία τ=η βα και τ=ω πρα)υτ=η σημείω τ=ω α δ/υο ευθε=ιαι αι αγ, αη μ)επτα)υτμ/ερη
 κείμεναι τ)εφεχ=ηξ γων/ιαξ δυς)ορθάιξ)/ισαξ ποιο=υσιν;)επ) ευθείαξ)/αρα)εστ(η
 γα τ=η αη. δια τα)υτδκαι (η βα τ=η αθ)εστιν)επ) ευθείαξ. και)επει)/ιση)εστ(η
 (υπδβγ γων/ια τ=η (υπζβγ;)ορθγ(εκατ/ερα; κοινπροσκέισθω (η (υπαβγ; (/ολη)/αρα
 (η (υπδβα (/ολη τ=η (υπζβγ)εστιν)/ιση. και)επει)/ιση)εστ(η μδβ τ=η βγ, (η δζβ
 τ=η βα, δ/υο δαι δβ, βα δ/υο ταιξ ζβ, βγ)/ισαι είς(εκατ/ερα (εκατ/ερα; και γων/ια
 (η (υπδβα γων/ια τ=η (υπζβγ)/ιση; β/ασιξ)/αρα (η αδ β/ασει τ=η ζγ [)εστιν])/ιση,
 και ταβδ τρ/ιγωνον τ=ω ζβγ τριγ/ωνω)εστ)/ισον; και [)εστι] το=υ μαβδ τριγ/ωνου
 διπλ/ασιον τβλ παραλληλ/ογραμμον; β/ασιν τε γτα)υτ)/εθουσι τβδ και)εν ταιξ α)υταιξ
 είσι παραλληλ/οιξ ταιξ βδ, αλ; το=υ δζβγ τριγ/ωνου διπλ/ασιον τηβ τετρ/αγωνον;
 β/ασιν τε γπ/αλιν τα)υτ)/εθουσι τζβ και)εν ταιξ α)υταιξ είσι παραλληλ/οιξ ταιξ ζβ, ηγ.
 [τδτ=ων)/ισων διπλ/ασια)/ισα)αλλ/ηλοιξ)εστ/ιν;])/ισον)/αρα)εστκαι τβλ παραλληλ/ογραμμον
 τ=ω ηβ τετραγ/ωνω. (ομοίωξ δ)επιζευνυμ/ενων τ=ων αε, βκ δειθθ/ησεται και τγλ
 παραλληλ/ογραμμον)/ισον τ=ω θγ τετραγ/ωνω; (/ολον)/αρα τβδεγ τετρ/αγωνον δυστο=ιξ
 ηβ, θγ τετραγ/ωνοιξ)/ισον)εστ/ιν. και)εστι τμβδεγ τετρ/αγωνον)απτ=ηξ βγ)αναγραφ/εν,
 τδηβ, θγ)απτ=ων βα, αγ. τ)/αρα)απτ=ηξ βγ πλευρ=αξ τετρ/αγωνον)/ισον)εσtto=ιξ
)απτ=ων βα, αγ πλευρ=ων τετραγ/ωνοιξ.

)εν)/αρα το=ιξ)ορθογων/ιοιξ τριγ/ωνοιξ τ)απτ=ηξ τ)ορθγων/ιαν (υποτεινόνσηξ
 πλευρ=αξ τετρ/αγωνον)/ισον)εσtto=ιξ)απτ=ων τ)ορθ[γων/ιαν] περιεθους=ων πλευρ=ων
 τετραγ/ωνοιξ; (/οπερ)/εδει δε=ιχαί.

Proposition 47

In right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides containing the right-angle.

Let ABC be a right-angled triangle having the angle BAC a right-angle. I say that the square on BC is equal to the (sum of the) squares on BA and AC .

For let the square $BDEC$ be described on BC , and (the squares) GB and HC on AB and AC (respectively) [Prop. 1.46]. And let AL be drawn through point A parallel to either of BD or CE [Prop. 1.31]. And let AD and FC be joined. And since angles BAC and BAG are each right-angles, then two straight-lines AC and AG , not lying on the same side, make the adjacent angles with some straight-line BA , at the point A on it, (whose sum is) equal to two right-angles. Thus, CA is straight-on to AG [Prop. 1.14]. So, for the same (reasons), BA is also straight-on to AH . And since angle DBC is equal to FBA , for (they are) both right-angles, let ABC be added to both. Thus, the whole (angle) DBA is equal to the whole (angle) FBC . And since DB is equal to BC , and FB to BA , the two (straight-lines) DB , BA are equal to the two (straight-lines) CB , BF ,[†] respectively. And angle DBA (is) equal to angle FBC . Thus, the base AD [is] equal to the base FC , and the triangle ABD is equal to the triangle FBC [Prop. 1.4]. And parallelogram BL [is] double (the area) of triangle ABD . For they have the same base, BD , and are between the same parallels, BD and AL [Prop. 1.41]. And square GB is double (the area) of triangle FBC . For again they have the same base, FB , and are between the same parallels, FB and GC [Prop. 1.41]. [And the doubles of equal things are equal

to one another.][‡] Thus, the parallelogram BL is also equal to the square GB . So, similarly, AE and BK being joined, the parallelogram CL can be shown (to be) equal to the square HC . Thus, the whole square $BDEC$ is equal to the (sum of the) two squares GB and HC . And the square $BDEC$ is described on BC , and the (squares) GB and HC on BA and AC (respectively). Thus, the square on the side BC is equal to the (sum of the) squares on the sides BA and AC .

Thus, in right-angled triangles, the square on the side subtending the right-angle is equal to the (sum of the) squares on the sides surrounding the right-[angle]. (Which is) the very thing it was required to show. [†] The Greek text has “ FB, BC ”, which is obviously a mistake.

[‡] This is an additional common notion.

48.

ἔαν τριγ/ωνου τ)απμῖαξ τ=ων πλευρ=ων τετρ/αγωνον)/ισον)=η το=ιξ)απτ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων τετραγ/ωνοιξ, (η περιεθομ/ενη γων/ια (υπτ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων)ορθ/η)εστιν.

τριγ/ωνου γτο=υ αβγ τ)απμῖαξ τ=ηξ βγ πλευρ=αξ τετρ/αγωνον)/ισον)/εστω το=ιξ)απτ=ων βα, αγ πλευρ=ων τετραγ/ωνοιξ; λ/εγω, (/οτι)ορθ/η)εστιν (η (υπβαγ γων/ια.

)ηθθω γ)απτο=υ α σημείου τ=η αγ ἐυθεία πρ)ορθ(η αδ καὶ κείσθω τ=η βα)/ιση (η αδ, καὶ)επεζεύθθω (η δγ.)επεί)/ιση)εστ(η δα τ=η αβ,)/ισον)εστκαὶ τ)απτ=ηξ δα τετρ/αγωνον τ=ω)απτ=ηξ αβ τετραγ/ωνω. κοινπροσκέισθω τ)απτ=ηξ αγ τετρ/αγωνον; τ)/αρα)απτ=ων δα, αγ τετρ/αγωνα)/ισα)εστω=ιξ)απτ=ων βα, αγ τετραγ/ωνοιξ.)αλλτο=ιξ μ)απτ=ων δα, αγ)/ισον)εστω)απτ=ηξ δγ;)ορθγ/αρ)εστιν (η (υπδαγ γων/ια; το=ιξ δ)απτ=ων βα, αγ)/ισον)εστω)απτ=ηξ βγ; (υπ/οκειται γ/αρ; τ)/αρα)απτ=ηξ δγ τετρ/αγωνον)/ισον)εστω=ω)απτ=ηξ βγ τετραγ/ωνω; (/ωστε καὶ πλευρ(η δγ τ=η βγ)εστιν)/ιση; καὶ)επεί)/ιση)εστ(η δα τ=η αβ, κοινδ(η αγ, δ/υο δαί δα, αγ δ/υο τᾶιξ βα, αγ)/ισαι εἰς/ιν; καὶ β/ασιξ (η δγ β/ασει τ=η βγ)/ιση; γων/ια)/αρα (η (υπδαγ γων/ια τ=η (υπβαγ [εστιν])/ιση.)ορθδ(η (υπδαγ;)ορθ)/αρα καὶ (η (υπβαγ.

ἔαν αρτριγ/ωνου τ)απμῖαξ τ=ων πλευρ=ων τετρ/αγωνον)/ισον)=η το=ιξ)απτ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων τετραγ/ωνοιξ, (η περιεθομ/ενη γων/ια (υπτ=ων λοιπ=ων το=υ τριγ/ωνου δ/υο πλευρ=ων)ορθ/η)εστιν; (/οπερ)εδει δε=ιχαί.

Proposition 48

If the square on one of the sides of a triangle is equal to the (sum of the) squares on the two remaining sides of the triangle (then) the angle contained by the two remaining sides of the triangle is a right-angle.

For let the square on one of the sides, BC , of triangle ABC be equal to the (sum of the) squares on the sides BA and AC . I say that angle BAC is a right-angle.

For let AD be drawn from point A at right-angles to the straight-line AC [Prop. 1.11], and let AD be made equal to BA [Prop. 1.3], and let DC be joined. Since DA is equal to AB , the square on DA is thus also equal to the square on AB .[†] Let the square on AC be added to both. Thus, the (sum of the) squares on DA and AC is equal to the (sum of the) squares on BA and AC . But, the (square) on DC is equal to the (sum of the squares) on DA and AC . For angle DAC is a right-angle [Prop. 1.47]. But, the (square) on BC is equal to (sum of the squares) on BA and AC . For (that) was

assumed. Thus, the square on DC is equal to the square on BC . So side DC is also equal to (side) BC . And since DA is equal to AB , and AC (is) common, the two (straight-lines) DA , AC are equal to the two (straight-lines) BA , AC . And the base DC is equal to the base BC . Thus, angle DAC [is] equal to angle BAC [Prop. 1.8]. But DAC is a right-angle. Thus, BAC is also a right-angle.

Thus, if the square on one of the sides of a triangle is equal to the (sum of the) squares on the remaining two sides of the triangle (then) the angle contained by the remaining two sides of the triangle is a right-angle. (Which is) the very thing it was required to show. [†] Here, use is made of the additional common notion that the squares of equal things are themselves equal. Later on, the inverse notion is used.